The Global AI Power Order

Research Supplement — Detailed Evidence and Adjudication Records

Companion to the v81 P2-33 editorial working manuscript | Research lineage: v79 master → v80 Supplement v1.0

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This independently archived supplement preserves detailed records omitted from the print appendices. Section identifiers S1–S6 provide stable citations; claim IDs, source wording, evidence ceilings, and negative/not-adjudicated decisions are preserved. The source references are to the 413-item References section of the accompanying book. Post-cutoff review dates indicate verification activity only and do not backfill new outcomes into the 30 September classifications.

Contents and original appendix crosswalk

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Supplement S1 — Targeted Review and Dynamic-Source Evidence

Extended cutoff-sensitive evidence checks and preserved post-cutoff retrieval extracts from original Appendix A.8–A.10. These are research audit records, not additional historical observations.

S1.1 Targeted Status Review Records

A targeted status review conducted on 5 October 2026 applied this procedure to three cutoff-sensitive records. A subsequent New Delhi assessability re-read was recorded on 6 October 2026. These review dates are later than the empirical cutoff and are used only to verify the state that existed on or before 30 September 2026; post-cutoff developments are not backfilled into the classification. The original three records and the subsequent New Delhi record are kept as distinct versioned entries.

STAT-ANTH-20261005

Field	Record
Object / source	D.C. Circuit Anthropic military-contract opinion
Source date	25 Sep 2026
Review date	5 Oct 2026
Cutoff-state question	Was the military-contract restriction operative by cutoff?
Cutoff finding	Yes at the specified Pentagon military-contract interface; broader federal restrictions remain separately bounded.
Stopping rule	Dated appellate opinion directly resolves the cutoff status for the specified interface.

STAT-SG-DIB-20261005

Field	Record
Object / source	Singapore Digital Infrastructure Bill / MDDI First Reading notice
Source date	8 Sep 2026
Review date	5 Oct 2026
Cutoff-state question	Was the proposed bill already an enacted/operational licensing regime?
Cutoff finding	No. First Reading / proposed legislative stage at cutoff; coded as institutional development, not operative licensing.
Stopping rule	Dated official First Reading notice fixes legislative stage.

STAT-MY-AIM-20261005

Field	Record
Object / source	AI Malaysia Berhad institutional status
Source date	28 Jul & 11 Sep 2026
Review date	5 Oct 2026
Cutoff-state question	Was AI Malaysia Berhad institutionally established by cutoff?
Cutoff finding	Yes. Official establishment announced 28 Jul; inaugural board announced 11 Sep. Planned 2030 capabilities remain future trajectory.
Stopping rule	Two dated Ministry of Digital releases establish institution and governance before cutoff.

STAT-ND-20261006 — New Delhi commitment assessability re-read

The v25 review rechecked the dated Government of India commitment record and the pre-cutoff Anthropic transparency record. The latter documents selected follow-through but also places at least one commitment-specific deliverable at the next AI Summit, so the manuscript does not infer that the entire 13-provider commitment set had reached a common due-and-observable implementation window by 30 September 2026. The SI candidate is therefore Not Adjudicated rather than Class 0; this is an assessability correction, not a finding of implementation failure.

S1.2 Dynamic-Source Retrieval Extract Register

For a small set of cutoff-sensitive dynamic sources that materially support current classifications, the v21 research package preserves retrieval extracts created on 5 October 2026. Each extract records the source URL, retrieval date, cutoff-relevant observation, analytical role, evidentiary limit, and a SHA-256 hash. These files are research retrieval extracts rather than WARC captures or complete HTML mirrors; they preserve the adjudication-relevant state visible at retrieval and do not convert post-cutoff page content into proof of the 30 September 2026 state. The companion archive manifest is the authoritative index for these Archive IDs.

Archive ID	Source, Preserved Observation, and Evidentiary Limit
ARCH-FR-DINUM-20261005	DINUM — Assistant usage statistics; retrieved 5 Oct 2026. Preserves the displayed monthly record of 38,038 active Assistant users via ProConnect for 30 Sep 2026. Used for ANP-FR-PA-01; dynamic service pages do not expand the claim beyond the dated record.
ARCH-IN-DRDO-AIML-20261005	DRDO — AI/ML technology-task page; retrieved 5 Oct 2026. Preserves listed AI/ML development tasks. Used as institutional/technical-base evidence for ANP-IN-DI-01; does not establish classified deployment scale or mission performance.
ARCH-IN-DRDO-AUTO-20261005	DRDO — Autonomous systems and robotics task page; retrieved 5 Oct 2026. Preserves listed autonomy/robotics tasks. Used at the public technical-base level for ANP-IN-DI-01; operational breadth remains partially observable.
ARCH-DE-BOSCH-SHOPFLOOR-20261005	Bosch — Shopfloor Agent deployment page; retrieved 5 Oct 2026. Preserves Bosch-reported operational manufacturing use and external-customer availability since late 2025. Company-reported performance is not treated as independently verified national effect.
ARCH-DE-BOSCH-DRESDEN-20261005	Bosch — Dresden wafer-fab page; retrieved 5 Oct 2026. Preserves the described AIoT factory architecture and AI use in process analysis, defect detection, predictive maintenance, and optimization. Used as operational company evidence, not as a national adoption census.
ARCH-QA-KAHRAMAA-SMARTMETERS-20261005	KAHRAMAA — Smart Meters project page; retrieved 5 Oct 2026. Preserves architecture and stated deployment scale, including more than 500,000 smart electricity meters by end-2024. Live counts are not backfilled past the dated cutoff evidence.
ARCH-MY-AIM-FAQ-20261005	AI Malaysia Berhad — National AI Action Plan FAQ; retrieved 5 Oct 2026. Preserves mandate and plan-content context. Dated Ministry releases, rather than the undated FAQ alone, carry the cutoff-specific establishment and board-timing claims.

Supplementary archive package: Global_AI_Power_Order_v21_Dynamic_Source_Retrieval_Extracts.zip (SHA-256: 2d4773e3bf3fda6f062b0a20839b1b25214bea0c7c89b485d3513e3d0f2ae906).

Manifest: archive_manifest.csv (SHA-256: 576462ffde11df08d2ff2c44b13eafa166112b4ed292a339e569db6d75d6ad53); archive_manifest.json is included alongside it.

The package contains seven text extracts named by Archive ID. Each manifest record supplies the relative filename, source URL, retrieval date, preserved observation, evidentiary limit, and file hash. During the 6 October 2026 delivery reconciliation, the companion ZIP and CSV manifest were re-opened in the working environment: the ZIP contained the seven named extracts plus CSV and JSON manifests, and the recomputed ZIP and CSV SHA-256 values matched the values printed above. This verifies delivery-package byte integrity in the working environment; it does not convert 5 October retrieval extracts into pre-cutoff web snapshots or independently prove past webpage states.

Supplement S2 — Core Evidence Register and Adjudication Record

The full evidence register is designed as a living research record linking claims, tables, and maps to source-level observations. This appendix provides the schema and a core set of high-impact adjudications that were central to the September 2026 revision. It is not a substitute for the complete source corpus and should not be read as a sample from which frequencies are inferred.

B.1 Evidence Register Schema

Field	Purpose
Claim / Object ID	Stable identifier connecting narrative, table, map, and adjudication record.
Construct / unit / function	Specifies ANP, ABP, Boundary Autonomy, Systemic Influence, or higher-order structure and the correct analytical unit.
Observation window and dates	Separates event, publication, effective, implementation, observation, and retrieval dates.
Source and source role	Identifies the exact source and what type of proposition it can directly establish.
What is established	States the narrow claim directly supported by the observation set.
What remains unestablished	Records missing functional continuity, causal attribution, breadth, persistence, or other limits.
Contrary evidence / rival interpretation	Preserves observations that weaken or qualify the preferred interpretation.
Threshold and adjudication	Records the gate applied and the resulting time-indexed classification.
Map / table / version links	Connects the record to displayed synthesis objects and preserves revision history.

B.1.1 Observation-Pointer Convention

Each core adjudication row has a stable Claim/Object ID. Within the Evidence Basis and Timing field, discrete source-backed observations are read as claim-specific observation pointers in the form ClaimID/O1, ClaimID/O2, and so forth, in the order listed. The source key, date or observation period, and narrow empirical content identify the observation; the canonical URL is supplied by the corresponding References entry, with chapter or table locators and legal pinpoints added where available. Reuse of one source for more than one proposition does not merge the propositions: each observation pointer supports only the narrow statement stated in its row. This convention makes source role, timing, threshold relevance, and competing explanations auditable without duplicating full URLs across the core table.

B.1.2 Core Observation Source-Locator Bridge

The bridge below is deliberately selective. It covers observations that decide current BA, SI, or domain judgments and provides a stable source/version locator without copying long URLs into the register. Event time, publication time, and retrieval time remain distinct. Analytical synthesis stays in G records rather than being recoded as an observation.

OBS ID / Gate Use	Source / Version / Locator
OBS-DOM-US-SA-01 — strict same-function operational bridge	Advanced Micro Devices, Inc. (2026), 31 Aug. “AMD, Cisco and HUMAIN Expand…”; News Highlights and first deployment paragraphs. TS-C = AMD Instinct/EPYC compute; TS-N = Cisco Silicon One/N9000 fabric. Future 2027 expansion is excluded from cutoff capability.
OBS-DOM-US-NATO-01 — external domain context only	NATO Joint Warfare Centre (2026), 12 Aug.; opening paragraphs report MSS full technical operational capability achieved 22 Jun. NATO holds the accreditation/governance role; the record is not coded as U.S. control.
OBS-BA-CN-KERNEL-01 — cross-platform execution anchor	DeepSeek-AI (2026), arXiv:2606.19348v1; Version 1, cross-platform V4 expert-parallel validation passage. Supports a specified execution alternative, not national-scale availability.

OBS ID / Gate Use	Source / Version / Locator
OBS-BA-CN-PT-01 — post-training workload	Li et al. (2026), arXiv:2607.20145v3; Version 3, abstract and experiment/results discussion reporting full-parameter V4-family post-training on Ascend SuperPOD, 34.22% MFU, training stability, and CPT/SFT workflow. Author-reported technical implementation.
OBS-BA-CN-INF-01 — inference workload and limits	Yu (2026), arXiv:2607.08215v1; Version 1, abstract and field-study workload/results sections. Two V4-family inference workloads were served correctly on 16 Ascend 910 devices, with twelve patches and material throughput/reliability constraints.
OBS-SI-DPC-META-01 — bounded persistent supervisory change	Data Protection Commission (2025), 21 May; statement paragraphs describing the 2024 pause, revised 2025 proposal, and recommendations before launch. Supports the Meta relation, not provider-class breadth.
OBS-SI-DPC-AI-2026-01 — wider recurrence context	Data Protection Commission (2026), 24 Sep.; AI Insights Report summary: supervision of approximately 180 AI products/services during 2021–2025. Used as recurrence context only.
OBS-SI-UPI-START-01 — structural start point	Association of Banks in Singapore (2023), 21 Feb.; PayNow–UPI Linkage Fact Sheet, p. 1. Establishes launch and bilateral gateway architecture.
OBS-SI-UPI-CONT-01 — later persistence point	Monetary Authority of Singapore (2026), 11 Sep.; Global FinTech Fest address, paras. 51–53. States the linkage remained operative and transaction volumes had more than doubled each year since launch. Supports bilateral persistence, not broader diffusion.
OBS-ADV-* — interface-level advanced-compute changes	Use Table 13.2 relation IDs as the manuscript locator: A1/A2 accelerator, B1 DUV/service, C1 EDA, D1 HBM, E1 foundry. Each observation is tied to its cited regulatory/supplier source in Chapter 13; the aggregate Systemic Influence claim remains Not Adjudicated under the v48 system-relative breadth test.
OBS-BA-IMOD-DISABLE-01 — managed-service disablement	Microsoft (2025), 25 Sep., “Update on ongoing Microsoft review”; passage stating that specified subscriptions and services, including cloud-storage and AI-related services, were ceased and disabled. Supports an observed service-side boundary action, not completed target substitution.
OBS-BA-IMOD-SWITCH-01 — reported alternative pathway	Microsoft (2026), “Summary of 2025 external investigation and follow up”; passage summarizing public reporting that affected data or workloads might transfer to a competing cloud platform. Supports target response / planned switching only; migration completion and preserved function remain unestablished.
OBS-DOM-EU-DPC-01 — implemented EU/EEA supervisory recurrence	Data Protection Commission (2025), 21 May, Meta AI statement; and Data Protection Commission (2026), 24 Sep., AI Insights Report summary. These records support implemented and recurrent supervision inside the EU/EEA rule space; they do not establish diffusion beyond it.
OBS-ADV-EDA-HIST-01 — historical EDA restriction / rescission	Cadence Design Systems, Inc. (2025), May–2 Jul. sequence described in Ch.13 §13.3 and Ch.18 §18.5: the specified broad EDA licensing requirement reduced deliveries and was rescinded on 2 July 2025, after which access was restored. It is retained as a historical boundary episode, not counted as an active cutoff restriction.

B.2 Core September 2026 Adjudication Records

Claim / Object	Evidence, Adjudication, and Limits

Claim / Object	Evidence, Adjudication, and Limits
ID — ANP-UAE-RES-01 UAE stress-and-continuity evidence; national system × critical AI infrastructure × Sep. 2026	**Evidence basis and timing —** O1 — Reuters (2026l), 11 Sep. 2026: post-attack redesign toward geographic dispersion, hardening, protected siting, and some underground construction. Ch.11 §11.3; Ch.12 comparative synthesis. **Adjudication —** Observed stress-responsive architectural adaptation established; severe-disruption resilience remains unestablished. **Limits / rival interpretation —** Redesign after a real shock is stronger than anticipatory planning, but it does not demonstrate continuity under a comparable future disruption.
ID — ABP-APPLE-CN-01 CAC / relevant Chinese GenAI authorities → Apple / Apple Intelligence; public-facing GenAI market-entry interface	**Evidence basis and timing —** O1 — Apple (2026c), 14 Sep. 2026: new features launched elsewhere while remaining unavailable in China pending regulatory requirements. O2 — CAC filing evidence: the applicable market-entry / filing condition. O3 — Reuters (2026i): external reporting on the China-specific availability constraint. Ch.15 §15.10 and Table 15.2. **Adjudication —** State 2 operational boundary control established at the specified market-entry interface; observable State 3 target adaptation. **Limits / rival interpretation —** No named CAC denial or suspension order is inferred; the claim concerns operationally consequential admission conditions, not a formal ban.
ID — ABP-ANTH-DOD-01 U.S. Department of War / Pentagon → Anthropic; military-contract participation interface	**Evidence basis and timing —** O1 — D.C. Circuit opinion, 25 Sep. 2026: military-contract participation restriction and appellate disposition (see Ch.16 §16.14 for pinpoints). O2 — earlier district-court record: preceding procedural and scope context. O3 — Reuters (2026m): contemporaneous reporting on implementation and litigation. Ch.16 §16.14. **Adjudication —** Active State 2 restrictive boundary established at the Pentagon military-contract interface as of 30 Sep. 2026. **Limits / rival interpretation —** Do not generalize to company-wide federal exclusion; broader restrictions depended on separate legal authorities and litigation.
ID — BA-RUS-COMPUTE-01 Russia; alternatives relative to constrained Western advanced-compute interfaces	**Evidence basis and timing —** O1 — Government of the Russian Federation (2026c), Higher School of Economics (2026), Sber (2026), and Yandex (2025): source-specific evidence of continuing downstream model, platform, administrative, industrial, and scientific functions under upstream constraint (Ch.10 §§10.5–10.10). O2 — TOP500 (2026a, 2026b, 2026c, 2026d, 2026e): continued listing of legacy GPU-based systems, not independently verified utilization. O3 — Reuters (2026e), 20 May, and YADRO (2026), 9 Feb.: alternative sourcing and domestic server-integration pathways. G1 — Ch.17 §17.2 applies the BA gate; the adjudication is not counted as an observation. **Adjudication —** Adaptation is observable; Boundary-Autonomy gain remains insufficiently established for the specified function. **Limits / rival interpretation —** Foreign sourcing is not disqualifying. Unresolved issues are operational availability, functional continuity, switching cost, durability, and independence from the original controlled interface.

Claim / Object	Evidence, Adjudication, and Limits
ID — BA-IMOD-CLOUD-01 IMOD target response at managed-cloud / remote-administration interface	Evidence basis and timing — O1 / OBS-BA-IMOD-DISABLE-01 — Microsoft (2025, 25 Sep.), "Update on ongoing Microsoft review": Microsoft ceased and disabled specified subscriptions and services, including named cloud-storage and AI-related services, at the specified Israel Ministry of Defense interface. O2 / OBS-BA-IMOD-SWITCH-01 — Microsoft (2026), "Summary of 2025 external investigation and follow up": the follow-up summary notes public reporting that affected data or workloads might move to a competing cloud platform. Ch.14 Table 14.3. Adjudication — Alternative pathway identified and movement initiated or planned; realized Boundary-Autonomy gain not established. Limits / rival interpretation — Planning or migration start does not establish completed migration, equivalent workload performance, or sustained operation.
ID — SI-MOSIP-01 Cross-national modular digital-identity architecture; adjacent DPI calibration case	**Evidence basis and timing —** O1 / OBS-SI-MOSIP-PH-01 — PSA (2024): PhilSys identifies MOSIP lineage and national adaptation. O2 — PSA (2026): continuing authentication use in an operating adopter system. O3 / OBS-SI-MOSIP-ET-01 — GCS (2026): Fayda operating scale. O4 — MOSIP project records: architecture-lineage and implementation-stage evidence. G1 — Ch.19 population review: the full cutoff-date engagement cohort is not independently enumerated by stage. **Adjudication —** Not Adjudicated for Systemic Influence. Selected implementations support structural change, persistence, and architecture lineage, but population-level breadth cannot be validly assessed from a success-filtered set. **Limits / rival interpretation —** Adjacent DPI calibration only; not AI-specific or India-wide SI. Continuing operation and lineage are outcomes/evidence, not population-entry conditions. A future Class 0–3 result requires a complete, independently defined cohort and stage register.
ID — SI-DPC-META-01 DPC–Meta AI-training supervisory outcome within relevant EU/EEA data-protection environment	Evidence basis and timing — O1 / OBS-SI-DPC-META-01 — Data Protection Commission (2025, 21 May), the Meta AI statement describing the 2024 pause, the revised 2025 proposal, and supervisory recommendations: implemented supervisory change in the specified Meta relation. O2 / OBS-SI-DPC-AI-2026-01 — Data Protection Commission (2026, 24 Sep.), AI Insights Report: supervision of approximately 180 AI products/services during 2021–2025; used as wider recurrence context, not as attribution rescue. Ch.19 §§19.2, 19.14 and Table 19.3. Adjudication — Not Adjudicated for the provider-class systemic claim; persistent change is established in the Meta relation, but a valid wider breadth assessment is unavailable. Limits / rival interpretation — the wider supervision record does not establish that the Meta-specific outcome recurred across independent providers because of the same intervention.

Claim / Object	Evidence, Adjudication, and Limits
ID — DOMAIN-US-01 U.S.-Centered AI Domain; higher-order technological/institutional structure	Evidence basis and timing — O1 — Abu Dhabi Media Office (2026) and Core42 (2026): UAE government-wide Microsoft enterprise-AI deployment and separately embedded sovereign accelerator access; same-workload integration not established. O2 / OBS-DOM-US-SA-01 — Advanced Micro Devices, Inc. (2026), 31 Aug.: HUMAIN production AI compute live and serving customers. Strict-test layer coding: TS-C = AMD Instinct MI355X / EPYC compute; TS-N = Cisco Silicon One / N9000 networking fabric; both operate inside the same production AI-infrastructure function. O3 — Ooredoo Group (2025): NVIDIA-accelerated sovereign infrastructure in Qatar; retained as single-layer recurrence context. O4 / OBS-DOM-US-NATO-01 — NATO JWC (2026), technical capability achieved 22 Jun. and reported 12 Aug.: Maven Smart System operates on NATO's classified network and supports multi-origin models; NATO, not the United States, holds the security accreditation. / Adjudication — baseline domain Supported as a recurrent non-exclusive pattern. Strict same-function/two-U.S.-origin-layer sensitivity Partially Supported: Saudi Arabia supplies one clear TS-C↔TS-N operational bridge, but repeated qualifying bridges across the external domain are not established. / Limits / rival interpretation — layer identity is analytic and function-specific, not supplier counting. NATO governance, Qatar single-layer use, and UAE environment-level coexistence are not promoted into additional strict passes.
ID — DOMAIN-CN-01 China-Centered External AI Configuration	**Evidence basis and timing —** O1 — external China-origin model uptake documented in Chs.6 and 19. O2 — selected foreign organizational use documented in Ch.20 §20.4. O3 — industrial/infrastructure relations in Chs.13–16. O4 — September OpenRouter evidence: external transaction activity, treated as reach evidence rather than durable-domain proof. **Adjudication —** PARTIALLY SUPPORTED external configuration; external domain coherence not established. **Limits / rival interpretation —** External reach is substantial, but durable cross-layer organization of foreign systems around one China-centered architecture remains incomplete.
ID — DOMAIN-EU-01 EU AI-Relevant Regulatory Domain	Evidence basis and timing — O1 / OBS-DOM-EU-DPC-01 — Data Protection Commission (2024; 2025; 2026): implemented supervisory intervention and recurrent AI supervision inside the originating EU/EEA rule space. O2 — EU institutional procurement and market-access records in Chs.15 and 20: recurring participation conditions. O3 — digital-service and data-use conditions in Chs.15, 19, and 20: additional rule-layer recurrence. Adjudication — SUPPORTED recurrent multi-state rule domain. Limits / rival interpretation — not a unified European technology stack and not evidence that EU AI regulation has already diffused globally beyond the controlled market; SI-EU-EXT-01 remains Not Adjudicated.

Claim / Object	Evidence, Adjudication, and Limits
ID — SI-ADVCOMP-01 China-related advanced-compute access-and-compliance system; scope-locked semiconductor/computing access domain	Evidence basis and timing — O1 / OBS-ADV-ACCEL-01 — accelerator access and supplier implementation in Table 13.2 A1/A2. O2 / OBS-ADV-DUV-01 — documented Dutch DUV controls in B1. O3 / OBS-ADV-EDA-HIST-01 — the May–2 July 2025 broad EDA licensing requirement was rescinded and access restored; this is historical, not a cutoff restriction. O4 — HBM, foundry, Japan-specific, remote-compute, and Malaysian relations vary in implementation and China-specific relevance. G1 — observed changes recur across interfaces, but the selected diagnostic cases are not an independently assessed census or representative population of China-related advanced-compute transactions and functions. Adjudication — Not Adjudicated: material structural change and temporal recurrence are observed; system-relative Sufficient Breadth is Unassessable and aggregate Credible Attribution is Not Reached. Limits / rival interpretation — The functional universe must include affected and unaffected, licensed and denied, restored and persistently restricted relations. The evidence does not establish absence of broader effects or a unified controller.
ID — SI-UPI-PAYNOW-01 Cross-border fast-payment interoperability architecture beyond the India–Singapore dyad	Evidence basis and timing — O1 / OBS-SI-UPI-START-01 — Association of Banks in Singapore (2023), PayNow–UPI Linkage Fact Sheet, p. 1: linkage launched in February 2023 and operated through cross-border gateways built and run by the two systems' operators. O2 / OBS-SI-UPI-CONT-01 — Monetary Authority of Singapore (2026, 11 Sep.), Global FinTech Fest address, paras. 51–53: UPI–PayNow remained an operating bilateral linkage and transaction volumes had more than doubled each year since launch. Adjudication — Not Adjudicated for the wider cross-border interoperability claim; persistence is established in the original bilateral linkage, but system-relative breadth cannot be validly tested. Limits / rival interpretation — persistence within one bilateral architecture does not count as independent recurrence across multiple systems.
ID — SI-NEWDELHI-01 Frontier-AI voluntary commitment implementation across the scope-locked signatory set	Evidence basis and timing — O1 — MeitY (2026a, 19 Feb.) and MeitY (2026b, 11 Mar.): the New Delhi commitment episode and 13-provider signatory count are observable. O2 — Anthropic (2026b), pre-cutoff transparency record: selected organizational follow-through is visible, while at least one commitment-specific deliverable (statistical insights) is framed for publication by the next AI Summit rather than by 30 Sep. 2026. G1 — the signatory population is enumerable, but the frozen record does not establish a common maturity window, commitment-specific due dates, or a signatory-wide negative search sufficient for an assessable Class 0. / Adjudication — Not Adjudicated for Systemic Influence. Endorsement reach and selected follow-through are established; recurrent structural implementation across the commitment set cannot be validly classified within the frozen window. / Limits / rival interpretation — This is not a finding that implementation failed. A later commitment-specific register could support Class 0–3 if due dates, observable outcomes, and signatory-level follow-up are independently specified.

Claim / Object	Evidence, Adjudication, and Limits
ID — SI-C2PA-01 AI-content provenance and trusted-recognition ecosystem	Evidence basis and timing — O1 / OBS-SI-C2PA-ARCH-01 — C2PA Conformance Program / Trust List: operational conformance and official recognition architecture. O2 / OBS-SI-C2PA-PROD-01 — dated conforming-product records available before the cutoff show more than one implementation; the official product-list schema distinguishes recordId, productType, conformance/public-disclosure dates, and status. G1 — these records establish operational recognition and multiple conforming products, but the frozen manuscript does not independently enumerate the cutoff-date product/platform population needed for a system-relative breadth test. Adjudication — Not Adjudicated at ecosystem level. Limits / rival interpretation — conforming-product plurality is not itself evidence of global ecosystem restructuring.
ID — SI-CNMODELS-01 Foreign organizational use of China-origin open / open-weight model ecosystems	**Evidence basis and timing —** O1 — MIIT (2026c): very large cumulative download reporting. O2 — Marchandon (2026): selected European organizational use. O3 — Mak (2026): OpenRouter transaction activity. G1 — these observations establish reach and selected use, but the current record does not independently enumerate an adopter population or a comparable before/after structural baseline for foreign stacks, workflows, or dependencies. **Adjudication —** Not Adjudicated for system-level SI. External uptake is established, but population-level structural change and persistence cannot be validly assessed from the current corpus. **Limits / rival interpretation —** Downloads and routing shares can reflect evaluation, repeated retrieval, or temporary multi-homing. Without a defined adopter population and longitudinal structural baseline, failure to observe durable reorganization cannot be coded as Class 0.
ID — SI-EU-EXT-01 AI-regulatory diffusion beyond the EU/EEA controlled market	**Evidence basis and timing —** O1 — EU/DPC supervisory evidence: implemented recurrence inside the originating EU/EEA rule space. O2 — market-access and participation-condition evidence in Chs.15 and 19. G1 — the externalization claim requires a separately defined off-market population and observable foreign replication/adaptation outcome; the current corpus does not enumerate that comparison population adequately. **Adjudication —** Not Adjudicated for AI-regulatory externalization beyond the EU/EEA controlled market. Internal domain recurrence remains separately supported. **Limits / rival interpretation —** Strong internal rule recurrence does not establish or refute off-market diffusion. Without a sufficiently defined external population and outcome window, the negative externalization claim is not assessable as Class 0.

Claim / Object	Evidence, Adjudication, and Limits
ID — BA-CN-ASCEND-01 China; specified DeepSeek-V4-family post-training and inference workloads relative to the U.S.-regulated Nvidia/CUDA-centered advanced-accelerator interface	Evidence basis and timing — O1 / OBS-BA-CN-KERNEL-01 — DeepSeek-AI (2026), arXiv:2606.19348v1, submitted 26 Apr.: V4 expert-parallel execution validated on NVIDIA GPUs and Huawei Ascend NPUs. O2 / OBS-BA-CN-PT-01 — Li et al. (2026), arXiv:2607.20145v3, pre-cutoff version: full-parameter DeepSeek-V4-family post-training on Ascend SuperPOD; 34.22% MFU, maintained training stability, CPT/SFT workflow. O3 / OBS-BA-CN-INF-01 — Yu (2026), arXiv:2607.08215v1, 9 Jul.: DeepSeek-V4-Flash MoE-judge and V4-Flash-Vision inference workloads served correctly on 16 Ascend 910 devices, with twelve source-level patches, disabled high-throughput features, and recurring-failure safeguards. Ch.17 §17.3. / Adjudication — specified operational alternative established for the source-documented post-training and inference workloads outside the original Nvidia-centered interface. A temporal Boundary-Autonomy gain is not separately established because the frozen record lacks a task-matched earlier comparison point. / Limits / rival interpretation — No same-condition Nvidia performance equivalence is inferred. Long-run production duration, economic cost, broad workload portability, HBM/fabrication/tooling depth, and severe-shock continuity remain incompletely observed; Yu (2026) documents substantial engineering and reliability constraints.

B.2.1 Scope, Gate, and Sensitivity Reconciliation Cards

These cards are retrospective review records for the v19 adjudication. Review-lock date means the date on which the scope was frozen for this revision; it is not a reconstructed preregistration date. Observation windows remain bounded by the 30 September 2026 empirical cutoff. Where the comparison population cannot yet be validly enumerated, the result is Not Adjudicated rather than a favorable or unfavorable class by default.

SI-MOSIP-01 — Scope/Gate Card

Field	Record
Review lock	5 Oct 2026; retrospective review-lock for v19.
Functional universe	Country-level MOSIP engagements within cross-national modular digital-identity architecture documented by the cutoff, irrespective of maturity stage.
Selection rule	Procurement, pilot, implementation, sustained operation, stalled, and other observable stages remain eligible population members; successful operation and traceable lineage are not entry conditions.
Observation window	Evidence through 30 Sep 2026; later retrieval may verify earlier dated states only.
Baseline / outcome	Baseline: pre-adoption national identity architecture. Candidate outcome: persistent adoption/adaptation of identifiable MOSIP architecture across independently governed national systems.
Gate result	Structural change and persistence supported in selected mature implementations; attribution supported in those cases; population-level breadth not validly assessable from the current success-weighted corpus.
Negative / rival case	Pilots or engagements without continuing operation remain in the population and cannot be silently excluded; local implementation capacity is a co-cause.
Current result	Not Adjudicated for Systemic Influence; retained as adjacent-DPI calibration of persistence/lineage/control separation.

SI-ADVCOMP-01 — Scope/Gate Card

Field	Record
Review lock	11 Oct 2026 v48 retrospective re-adjudication, superseding the 5 Oct v19 review lock; empirical evidence cutoff remains 30 Sep 2026.

Field	Record
Functional universe	Cross-border advanced-compute input and service relations relevant to China-based actors, including eligible transactions and functions regardless of whether controls were activated; accelerators, SME, EDA, HBM, foundries, remote compute, and relevant intermediary access.
Selection rule	The intended comparison includes authorized, denied, unaffected, restored, and rerouted relations as well as restrictive episodes. The assembled diagnostic register disproportionately observes implemented restrictions and selected reopenings; it has not characterized the full eligible comparison set.
Observation window	Repeated implementation from 2022 through 30 Sep 2026.
Baseline / outcome	Baseline: pre-2022 or pre-interface-change admissibility and function-specific supply relations as appropriate. Observed outcome: continuing but uneven changes in named licensing, due-diligence, verification, and participation relationships. A full population- or function-relative comparison has not been completed.
Gate result	Structural change and persistence in specified cross-interface relations are established. Relational escape and repeated implementation are supported, but system-relative breadth cannot be assessed adequately from an event-centered sample; credible aggregate attribution is not reached under the formal gate.
Negative / rival case	Selective licensing and reopening, historical EDA rescission, Japanese authorization without named denials, Dutch controls with narrow product coverage, and Malaysian enforcement without a confirmed China destination are retained as counterweights rather than folded into a uniformly restrictive aggregate.
Current result	Not Adjudicated (v48). Reopen Class 2 only after independently specifying the full comparison universe and demonstrating Sufficient Breadth; do not assign Class 1 simply because breadth is unproven.

SI-DPC-META-01 — Scope/Gate Card

Field	Record
Review lock	5 Oct 2026.
Functional universe	Meta AI-training data-use relation under relevant EU/EEA supervisory environment; broader provider recurrence treated separately.
Selection rule	Does not infer provider-class effects from the Meta dyad.
Observation window	2024–30 Sep 2026.
Baseline / outcome	Proposed training architecture → pause, safeguards/documentation redesign, and later continuation under revised arrangements.
Gate result	The initiating implementation exhibits a material, persistent change; an adequate independent comparison across the wider scope-locked population has not been completed, so system-relative breadth cannot be adjudicated.
Negative / rival case	Broader DPC supervision is context, not evidence that the Meta intervention caused provider-class restructuring.
Current result	Not Adjudicated (system-level SI claim); bounded initiating-relation persistence retained.

SI-UPI-PAYNOW-01 — Scope/Gate Card

Field	Record
Review lock	5 Oct 2026.
Functional universe	Cross-border fast-payment interoperability beyond the India–Singapore bilateral implementation.
Selection rule	Multiple endpoints inside one bilateral link do not count as independent recurrence.
Observation window	2023–30 Sep 2026.

Field	Record
Baseline / outcome	Separate payment rails → operational bilateral interoperability.
Gate result	The initiating implementation exhibits a material, persistent change; an adequate independent comparison across the wider scope-locked population has not been completed, so system-relative breadth cannot be adjudicated.
Negative / rival case	Transaction volume within the same link cannot substitute for independent recurrence across other systems.
Current result	Not Adjudicated (system-level SI claim); bounded initiating-relation persistence retained.

SI-NEWDELHI-01 — Scope/Gate Card

Field	Record
Review lock	5 Oct 2026 review lock; assessability correction recorded in the subsequent 6 Oct 2026 re-read. The later review date does not alter the 30 Sep 2026 empirical cutoff.
Functional universe	Frontier-AI voluntary commitment signatory set.
Selection rule	All signatories remain in the implementation assessment; endorsement does not count as implementation.
Observation window	19 Feb. 2026 commitment episode–30 Sep. 2026. This interval is not treated as proof that every commitment had reached a common due-and-observable implementation window.
Baseline / outcome	Voluntary endorsement → candidate recurring implementation changes.
Gate result	The signatory set is enumerable, but the frozen record does not establish that every commitment had reached a common due-and-observable implementation window by 30 September 2026. The collective implementation claim is therefore Not Adjudicated.
Negative / rival case	Selected follow-through is documented. An incomplete public record, uneven commitment-specific deadlines, and unobserved private changes do not establish implementation failure across the full signatory set. An assessable negative finding would require specified deliverables, eligible signatories, due dates, and observable outcomes.
Current result	Not Adjudicated. This replaces the earlier Class 0 judgment through an assessability correction; it is not a finding of implementation failure.

SI-C2PA-01 — Scope/Gate Card

Field	Record
Review lock	5 Oct 2026.
Functional universe	C2PA AI-content provenance / official conformance and trust-recognition ecosystem; cutoff-date comparison population must be defined independently of the products used as examples.
Selection rule	Conforming products may enter the cohort regardless of whether they support a positive SI finding; example products do not define the denominator.
Observation window	Mid-2025 program launch–30 Sep. 2026; current corpus lacks an independently frozen cutoff-date product roster with stage/outcome coding.
Baseline / outcome	Open specification and conformance framework → candidate recurring trusted-recognition architecture.
Gate result	Operational architecture and multiple dated implementations are established, but population-level persistence and breadth cannot be validly adjudicated from the current frozen record.
Negative / rival case	A one-implementation negative is no longer defensible because multiple conforming products are observable; the missing element is a frozen population and system-relative outcome test.
Current result	Not Adjudicated.

SI-CNMODELS-01 — Scope/Gate Card

Field	Record

Field	Record
Review lock	5 Oct 2026.
Functional universe	Foreign organizational use of China-origin open/open-weight model ecosystems; the relevant adopter population must be defined independently of high-visibility download or routing observations.
Selection rule	Downloads, routing shares and evaluations do not count as durable organizational embedding by themselves.
Observation window	Evidence through 30 Sep. 2026; organization-level start dates and comparable longitudinal follow-up are heterogeneous and incompletely enumerated.
Baseline / outcome	Model availability/uptake → candidate durable reorganization of foreign stacks, workflows or dependencies.
Gate result	Reach and selected use are established; a valid population-level test of structural change and persistence is not possible from the current corpus.
Negative / rival case	High downloads and routing activity are compatible with evaluation, temporary routing, or durable adoption. The current record does not distinguish these outcomes across a defined population.
Current result	Not Adjudicated.

SI-EU-EXT-01 — Scope/Gate Card

Field	Record
Review lock	5 Oct 2026.
Functional universe	AI-regulatory diffusion beyond the EU/EEA controlled market; external jurisdictions/firms must be defined independently of the internal EU cases used to establish the originating rule domain.
Selection rule	Internal EU recurrence does not count as externalization.
Observation window	Evidence through 30 Sep. 2026; the current corpus does not contain a sufficiently enumerated off-market comparison population with common follow-up.
Baseline / outcome	EU rules inside originating jurisdiction → candidate durable off-market replication or behavioral adaptation.
Gate result	Internal rule recurrence is established, but the externalization claim cannot be validly adjudicated for structural change, persistence, or breadth outside the originating market.
Negative / rival case	Absence of a documented external diffusion record in a selective corpus is not an assessable negative finding. A future test needs a defined off-market population and outcome window.
Current result	Not Adjudicated.

DOMAIN-US-01 — Common-Field Threshold Card

Field	Record
Scope	Non-exclusive U.S.-centered external AI domain across independently governed foreign functions.
Repeated connection	Saudi infrastructure supplies the clearest direct operational bridge: AMD compute plus Cisco networking in one HUMAIN infrastructure function (Advanced Micro Devices, Inc., 2026). NATO supplies platform-to-governed-network embedding in defense, but accreditation is NATO and model support is multi-origin (NATO JWC, 2026). Qatar is single-layer context; UAE same-workload integration is unestablished.
Persistence window	Operational relations observed through 30 Sep 2026; the Saudi production deployment is dated 31 Aug 2026 and NATO full technical operational capability is documented on 12 Aug 2026. Short-lived announcements and planned future capacity are excluded.
External effect	Baseline domain evidence shows recurring foreign functional embedding of U.S.-origin technologies. The stricter origin-layer test asks whether the same documented foreign function contains at least two separately identified U.S.-origin layers; it does not count two suppliers automatically as two layers.

Field	Record
Authority / attribution	Technological origin, controller, operator, and accreditation authority are recorded separately. HUMAIN operates the Saudi environment; NATO governs its classified network and security accreditation. No single U.S. controller is inferred.
Counterexample / limit	UAE separation, multi-homing, non-U.S. controllers, open models, and external rules limit bloc interpretation.
Result	Baseline result: Supported, non-exclusive, polycentric domain. Strict same-function / two-U.S.-origin-layer sensitivity: Partially Supported; only the Saudi record clearly survives as a direct operational bridge, while NATO, Qatar, and UAE are retained only at their narrower evidentiary roles.

DOMAIN-CN-01 — Common-Field Threshold Card

Field	Record
Scope	Foreign organizational configurations around China-origin models, infrastructure, industrial relations, and selected platforms.
Repeated connection	External model uptake and some organizational/industrial links recur, but cross-layer recurrence is uneven.
Persistence window	Contemporary reach is established; durable longitudinal embedding remains incomplete.
External effect	Foreign actors use China-origin models and selected infrastructure, often alongside U.S./European services.
Authority / attribution	Self-hosted/open-weight use reduces continuing Chinese provider control; state, firm and open-source roles remain separate.
Counterexample / limit	Multi-homing, self-hosting, and lack of common cloud–model–rule organization prevent a mature external domain judgment.
Result	Partially supported external configuration; domain not established.

DOMAIN-EU-01 — Common-Field Threshold Card

Field	Record
Scope	EU AI-relevant regulatory domain within the controlled market/rule space.
Repeated connection	Supervisory implementation, procurement/market-access rules, and data/service participation conditions recur across member-state/institutional settings.
Persistence window	Repeated implementation through the cutoff, not law-on-the-books alone.
External effect	Covered actors adapt participation, procurement, data-use or service conditions within the rule space.
Authority / attribution	EU/member-state institutional authorities and implementation bodies are distinguished; rule coherence does not imply a unified technology stack.
Counterexample / limit	External diffusion beyond the EU/EEA rule space is separately Not Adjudicated under SI-EU-EXT-01. Internal rule-domain support does not establish a global Brussels effect.
Result	Supported recurrent multi-state rule domain within bounded rule space.

BA-CN-ASCEND-01 — Workload Card

Field	Record
Function / workload	Two specified DeepSeek-V4-family workloads: (1) full-parameter post-training/CPT-SFT on Ascend SuperPOD for operations-research tasks (Li et al., 2026; arXiv:2607.20145v3); (2) inference serving for a DeepSeek-V4-Flash MoE judge and V4-Flash-Vision multimodal benchmark on 16 Ascend 910 devices (Yu, 2026; arXiv:2607.08215v1). DeepSeek-AI (2026; arXiv:2606.19348v1) supplies the cross-platform V4 execution-validation anchor.

Field	Record
Externally controlled interface	U.S.-regulated access to Nvidia-class advanced accelerators and the CUDA-centered deployment route; the claim is about an alternative to that interface, not elimination of all foreign dependencies.
Alternative path	Huawei Ascend 910 / Ascend SuperPOD with CANN/vLLM-Ascend, workload-specific kernel and systems optimization, and source-level adaptation.
Observed operation	Post-training: Li et al. (2026) report full-parameter V4-family post-training with 34.22% MFU, maintained training stability, and completed CPT/SFT workflow. Inference: Yu (2026) reports both specified workloads served correctly on the 16-device Ascend system.
Usability standard	Task-matched usability requires completed execution with acceptable task output and operational stability for the documented workload. Li et al. report maintained training stability and downstream task results; Yu reports correct serving and benchmark-quality evaluation. No inference is made to unrelated workloads or same-condition Nvidia equivalence.
Cost / continuity limits	Yu (2026) required twelve source-level patches, disabled several high-throughput features, and added safeguards for recurring device failures. Long-run production continuity, total cost, broad workload portability, frontier-scale availability, and upstream HBM/fabrication/tooling independence remain unestablished.
BA result	Specified operational alternative established for the documented post-training and inference workloads: the feasible route includes Ascend outside the original Nvidia-centered interface. A temporal Boundary-Autonomy gain is not separately established because the frozen record lacks a task-matched earlier comparison point. No full-stack, unrestricted frontier-pretraining, or cross-workload autonomy claim is made.

BA-RUS-COMPUTE-01 — Workload Card

Field	Record
Function / workload	No sufficiently specified advanced-compute workload with equivalent operational evidence is identified in the current public record.
Externally controlled interface	Constrained Western advanced-compute interfaces.
Alternative path	Alternative sourcing and downstream adaptation are observable.
Observed operation	Continuing downstream functions are observed, but the exact substitute path carrying a specified advanced workload is not adequately isolated.
Cost / continuity limits	Switching cost, performance, scale, durability, and independence from the original interface remain unresolved.
BA result	Adaptation observed; workload-level Boundary-Autonomy gain not established.

BA-IMOD-CLOUD-01 — Workload Card

Field	Record
Function / workload	Specified managed-cloud / AI service functions affected by provider disablement.
Externally controlled interface	Microsoft-managed service / remote-administration interface.
Alternative path	Reported/planned movement toward a competing cloud.
Observed operation	Alternative pathway identified; completed workload migration and sustained equivalent operation not established.
Cost / continuity limits	Performance, migration completion, switching cost and sustained continuity remain unobserved.
BA result	Realized Boundary-Autonomy gain not established.

Executed Sensitivity Results — v19–v26

v26 traceability reconciliation. The v25 assessability changes are retained: SI-NEWDELHI-01 and SI-EU-EXT-01 remain Not Adjudicated; BA-CN-ASCEND-01 retains task-level operational substitutability without a separately measured temporal gain; and DOMAIN-US-01 keeps the baseline Supported judgment while the strict sensitivity remains Partially Supported. v26 does not introduce a new classification. It adds source/version locators for the observations that decide these judgments, separates advanced-compute interface observations, and adds a pre-cutoff 11 September 2026 persistence anchor for UPI–PayNow.

Field	Record
MOSIP	Baseline v18: Class 3. Removed under the stricter test: success-filtered adopter sets, project adoption totals used as a denominator, and continuing operation / traceable lineage as population-entry conditions. Retained O IDs: OBS-SI-MOSIP-PH-01 and OBS-SI-MOSIP-ET-01 for selected-case persistence and lineage only. Gate change: population-level breadth cannot be assessed from the current cohort that has not been independently enumerated. Final result: Not Adjudicated; selected cases remain calibration evidence, not a population-level Class 3 result.
BA-CN	Baseline v18: bounded observed gain. Removed as constitutive BA evidence: shipment totals, announced orders, prospective collaborations, and generic pathway-development reports. Retained O IDs: OBS-BA-CN-KERNEL-01 (DeepSeek-AI, 2026; v1), OBS-BA-CN-PT-01 (Li et al., 2026; v3), and OBS-BA-CN-INF-01 (Yu, 2026; v1). Gate change: a task-matched operational alternative survives for specified post-training and inference workloads, while no comparable t0 is available for the same tasks. Final result: specified operational alternative established; temporal Boundary-Autonomy gain not separately adjudicated.
DOMAIN-US	Baseline v18: Supported. Removed from direct strict-bridge evidence: global prominence, Qatar single-layer hosting, UAE environment-level coexistence, and NATO accreditation treated as a U.S.-origin layer. Retained O ID: OBS-DOM-US-SA-01 (AMD/Cisco/HUMAIN live Saudi compute-network function). Context retained but not counted as a second strict bridge: NATO platform-to-governed-network embedding. Gate change: baseline recurrence survives, but repeated same-function/two-U.S.-origin-layer evidence does not. Final result: baseline domain Supported; strict origin-layer sensitivity Partially Supported.
ANP-TW-CI	Baseline label: Broad* commercial/industrial. Removed inference: that the single breadth label establishes equally broad domestic end-use across producer, enabler, and user roles. Retained evidence: semiconductor manufacturing, advanced packaging, ICT/server supply, and manufacturing-centered conversion records used for the producer/enabler scope. Gate change: role scope is narrowed, not the formal breadth label. Final result: Broad* retained only within the producer/enabler scope; no automatic inference to broad domestic downstream use.

B.3 Interpretation Rule

The register is not a hidden scorecard. An individual record can change without requiring neighboring constructs to change. For example, stronger Chinese accelerator alternatives can improve the evidence relevant to Boundary Autonomy without eliminating U.S. export-control ABP; wider DPC supervisory activity can strengthen evidence relevant to an EU rule domain without resolving the separate system-level breadth test for the DPC–Meta intervention; and selected MOSIP architectural adoption can remain an adjacent digital-identity observation without establishing AI-specific Systemic Influence. Versioned adjudication therefore updates only the objects whose evidentiary conditions have changed.

The core register includes a record for every candidate outcome in Table 19.3, together with the principal higher-order domain judgments and selected high-impact claims. The adjudication cards preserve the distinction between bounded implementation evidence and the separately assessed, scope-locked Systemic Influence claim.

Supplement S3 — Tier A ANP Comparative Adjudication Register

This appendix operationalizes the comparative matrix in Table 12.4 at the country–function level. Each Tier A cell is assigned a stable claim ID. Observed functional breadth, deployment maturity, and evidence sufficiency are reported separately so that a limit of public visibility is not silently converted into a judgment of weak underlying capability. The register points back to the relevant country chapter and sectoral profile. Supplement S4 then provides key source-level anchors, source roles, timing notes, and evidentiary ceilings for each of the 65 claim IDs. The entries are diagnostic and time-indexed to 30 September 2026. They are not components of an additive national score.

C.1 Coding and Reading Rule

Field	Meaning	Permitted inference	Prohibited inference
Observed Functional Breadth	Broad / Substantial / Selective where the public record permits a breadth judgment.	How widely operational use is observed within the specified function and window.	A cardinal power score or a claim of uniform performance.
Evidence Status	Partial / Insufficient where breadth cannot be validly adjudicated.	The limit of the public record for the specified function.	A claim that latent capability is low or absent.
Deployment Maturity	Descriptive state of implementation: operational, selected/bounded, expanding, mixed, trial/development, or not adjudicated.	How far the observed function has progressed from development toward recurring operation.	Automatic equivalence with breadth, effectiveness, resilience, or national benefit.
Evidence Sufficiency	Whether the record is sufficient for the stated breadth judgment.	Confidence that the breadth label itself is admissible.	A universal confidence score across countries or source types.

Boundary cases are resolved conservatively. A single factory deployment remains Selective even if technically sophisticated; recurring use across several major industrial subsectors can support Substantial, while Broad requires evidence that the pattern extends across multiple major components rather than merely several visible exemplars. A cross-agency public platform may support Substantial where it serves a nationally consequential set of tasks but still fall short of Broad if sustained use across materially different administrative functions is not demonstrated. For producer- or enabler-heavy systems, hardware production, compute provision, or training can support the role it actually performs, but it is not treated automatically as downstream end-user adoption. An upgrade in breadth therefore requires new observations crossing the relevant functional boundary, not simply more observations of the same narrow deployment.

The coverage unit differs by function. Commercial and Industrial uses firms, industrial subsectors, and production/service environments; Public Administration uses agencies, cross-agency platforms, and public-service tasks; Science and Education uses research institutions, allocable research infrastructure, scientific workflows, and education functions; Critical Infrastructure uses operators and infrastructure subsectors; Defense and Intelligence uses publicly observable mission, acquisition, support, and organizational functions. Because public visibility differs sharply, especially in defense and intelligence, evidence insufficiency is recorded separately rather than translated into a low-breadth judgment.

Observed Functional Breadth is adjudicated within each function using a function-appropriate coverage unit rather than a common national percentage. Selective means that operational evidence is confined to one or a small number of bounded organizations, sites, tasks, or subsectors. Substantial means that recurring operational evidence spans multiple consequential organizations, tasks, or subsectors, or a nationally pivotal function, but the record does not justify a broad claim across the function as a whole. Broad means that recurring operational evidence spans several major components of the specified function, or a dominant national layer through which a materially large share of that function is performed. These labels are ordinal only within the stated function and observation window; they are not cardinal power scores and are not mechanically comparable across functions. Role/Scope qualifier. Breadth labels apply only after the assessed role and function are specified. Broad producer or enabling capacity is not equivalent to broad domestic use; nationally embedded private operation is not automatically nationally mobilizable capacity. The role field therefore identifies whether the evidence primarily supports production, enabling infrastructure/access, end use, or a bounded combination of those roles. Where a country–function cell lists multiple roles, the breadth label applies to the combined role/function scope stated in that cell unless role-specific labels explicitly separate them; it does not imply that every listed role independently reaches the same breadth.

C.1.1 Breadth and Role Calibration Examples

The examples below show how the same ordinal vocabulary is applied across different evidence structures. The purpose is to make the Broad/Substantial/Selective boundary reusable rather than country-specific; no percentage threshold or hidden score is introduced.

Case / function	Role and component basis	Calibration result	Why the adjacent stronger/weaker label is not used

Case / function	Role and component basis	Calibration result	Why the adjacent stronger/weaker label is not used
United States — Science & Education	Producer + enabler + user; research compute, frontier/model research, universities/labs, national access/allocation pathways, and recurrent scientific/educational use	Broad	Multiple major components of the function are operationally connected; the judgment is not based on publications or talent alone.
South Korea — Commercial & Industrial	Producer + user; memory/semiconductor depth, electronics/manufacturing, platform/model activity, enterprise use, and industrial deployment	Broad	Breadth rests on several connected industrial components rather than one globally important input.
Taiwan — Commercial & Industrial	Producer + enabler; semiconductor manufacturing, advanced packaging, ICT/server supply, and manufacturing-centered conversion	Broad* within producer/enabler scope	The asterisk prevents upstream/system-supply breadth from being generalized to equally broad downstream services or end-use adoption.
Germany — Critical Infrastructure	User; live telecom-network AI is established while evidence across electricity, transport, health, and other subsectors is mixed	Selective	One strong operational subsector does not satisfy multi-subsector breadth for the composite critical-infrastructure function.
Qatar — Commercial & Industrial	Enabler + user; sovereign cloud and LNG-linked applications are operational but cross-sector evidence remains bounded	Selective	High strategic salience and infrastructure depth do not substitute for wider recurring conversion across the commercial/industrial function.
India — Commercial & Industrial	User + enabler; recurring operational AI across enterprise/services and selected industrial environments. Producer depth and upstream ownership are recorded separately and do not determine this breadth label.	Substantial	More than Selective because recurring operation spans multiple consequential user/enabler components within the commercial/industrial function. Below Broad because the frozen record does not establish similarly recurring coverage across several other major components of the same user/enabler functional scope. Compare South Korea Broad and Qatar Selective only as bounded calibration cases; producer depth, upstream ownership, and economy-wide maturity are separate limitations, not breadth penalties.

C.2 Claim Records by National System

C.2.1 United States

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-US-CI-01	Commercial & Industrial	Role / Scope: Producer + enabler + user. Observed Breadth: Broad	Operational at high maturity in frontier and large-firm environments; wider diffusion uneven	Sufficient for stated breadth judgment	Ch. 5 §5.12; Table 5.3. Exceptional resources and strong conversion; economy-wide diffusion remains uneven.
ANP-US-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational in selected agencies and functions; institutional maturity heterogeneous	Sufficient for stated breadth judgment	Ch. 5 §5.12; Table 5.3. Strong access to private technological base; procurement and implementation capability vary.

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-US-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Broad	Operational and strongly demonstrated	Sufficient for stated breadth judgment	Ch. 5 §5.12; Table 5.3. Exceptional research/compute depth and very strong public-private conversion; resilience remains untested comprehensively.
ANP-US-CR-01	Critical Infrastructure	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational across multiple subsectors; maturity sector-specific	Sufficient for stated breadth judgment	Ch. 5 §5.12; Table 5.3. Utility, telecom, transportation, and health functions are operational, but ownership and coordination remain decentralized.
ANP-US-DI-01	Defense & Intelligence	Role / Scope: User + enabler (publicly observable functions). Observed Breadth: Substantial	Operational in specified high-authority environments; broader integration partly classified	Sufficient for stated breadth judgment	Ch. 5 §5.12; Table 5.3. Dedicated acquisition and classified-network integration establish selected functions; public evidence does not expose the full system.

C.2.2 China

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-CN-CI-01	Commercial & Industrial	Role / Scope: Producer + enabler + user. Observed Breadth: Broad	Operational at scale; maturity uneven across firms	Sufficient for stated breadth judgment	Ch. 6 §6.12; Table 6.3. Very deep industrial base and strong conversion; upstream hardware and firm-level heterogeneity remain constraints.
ANP-CN-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational at multiple levels of government; regional maturity uneven	Sufficient for stated breadth judgment	Ch. 6 §6.12; Table 6.3. Provincial/municipal deployment and common-platform efforts establish use, but implementation quality varies.
ANP-CN-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Operational and substantial; AI-for-Science maturity uneven	Sufficient for stated breadth judgment	Ch. 6 §6.12; Table 6.3. Very deep research base and strong conversion; compute quality and application maturity vary by field.
ANP-CN-CR-01	Critical Infrastructure	Role / Scope: User + enabler. Observed Breadth: Broad	Operational across several major subsectors	Sufficient for stated breadth judgment	Ch. 6 §6.12; Table 6.3. Compute, electricity, telecom, and industrial infrastructures provide recurrent operational evidence; authority and safety constraints remain.
ANP-CN-DI-01	Defense & Intelligence	Role / Scope: User + enabler (publicly observable functions). Evidence Status: Partial	Selected support functions operational; combat/intelligence breadth not publicly adjudicated	Partial: breadth not adjudicated	Ch. 6 §6.12; Table 6.3. Institutional depth and support applications are visible; classification creates a hard evidence ceiling.

C.2.3 India

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-IN-CI-01	Commercial & Industrial	Role / Scope: User + enabler; producer role selective. Observed Breadth: Substantial	Operational across many downstream functions	Sufficient for stated breadth judgment	Ch. 7 §7.9; Table 7.3. Strong downstream and organizational depth; frontier compute, IP authority, and upstream ownership remain uneven.
ANP-IN-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Broad	Operational and strongly demonstrated	Sufficient for stated breadth judgment	Ch. 7 §7.9; Table 7.3. Institutional conversion through public digital and multilingual systems is well established; severe-shock continuity remains untested.
ANP-IN-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Operational/substantial and expanding	Sufficient for stated breadth judgment	Ch. 7 §7.9; Table 7.3. Research and educational use is growing, while frontier compute and tooling remain constraints.
ANP-IN-CR-01	Critical Infrastructure	Role / Scope: User (selected functions). Observed Breadth: Selective	Operational in selected functions; maturity uneven by sector	Sufficient for stated breadth judgment	Ch. 7 §7.9; Table 7.3. Selected infrastructure applications are established; broader sectoral continuity evidence is limited.
ANP-IN-DI-01	Defense & Intelligence	Role / Scope: Producer + user/enabler (public evidence partial). Evidence Status: Partial	R&D/platform development established; operational breadth partially observable	Partial: breadth not adjudicated	Ch. 7 §7.9; Table 7.3. Institutional and technological depth is visible, but classified deployment and operational breadth are not fully observable.

C.2.4 Japan

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-JP-CI-01	Commercial & Industrial	Role / Scope: Producer + user. Observed Breadth: Substantial	Operational and expanding; uneven across the wider economy	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Strong industrial conversion is visible, but breadth and maturity vary by sector.
ANP-JP-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational and rapidly scaling	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Government AI adoption is established and expanding; uniform maturity is not assumed.
ANP-JP-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Operational research/compute infrastructure	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Strong public compute and research infrastructure supports recurrent scientific use.

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-JP-CR-01	Critical Infrastructure	Role / Scope: User (selected functions). Observed Breadth: Selective	Selected operational deployment; mixed maturity	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Infrastructure applications are observable, but evidence is selective across subsectors.
ANP-JP-DI-01	Defense & Intelligence	Role / Scope: User/enabler (public evidence partial). Evidence Status: Partial	Not adjudicated beyond limited public evidence	Partial: breadth not adjudicated	Ch. 8; Table 8.3. Public evidence is insufficient for a comparable breadth judgment across defense and intelligence functions.

C.2.5 South Korea

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-KR-CI-01	Commercial & Industrial	Role / Scope: Producer + user. Observed Breadth: Broad	Broad operational manufacturing conversion	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Memory, electronics, manufacturing, and platform ecosystems support broad industrial use.
ANP-KR-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Government adoption is established in multiple functions.
ANP-KR-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Operational with strong model, compute, university, and corporate research base	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Research and model capacity are strongly evidenced across institutions.
ANP-KR-CR-01	Critical Infrastructure	Role / Scope: User + enabler. Observed Breadth: Broad	Operational in electricity and telecom	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Major infrastructure sectors show direct operational use.
ANP-KR-DI-01	Defense & Intelligence	Role / Scope: User/enabler (public evidence partial). Evidence Status: Partial	Not adjudicated beyond limited public evidence	Partial: breadth not adjudicated	Ch. 8; Table 8.3. Public evidence is insufficient for a comparable defense/intelligence breadth judgment.

C.2.6 Taiwan

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-TW-CI-01	Commercial & Industrial	Role / Scope: Producer + enabler; downstream user breadth not inferred. Observed Breadth: Broad*	Operational and mature in hardware/manufacturing-centered functions	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Semiconductor, ICT, server, and manufacturing strength supports broad commercial/industrial capability; this is not a claim of equally broad downstream services.
ANP-TW-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational and expanding	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Government adoption is established but still developing across functions.
ANP-TW-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Operational research/HPC and engineering depth	Sufficient for stated breadth judgment	Ch. 8; Table 8.3. Public HPC and semiconductor/engineering research are strong.
ANP-TW-CR-01	Critical Infrastructure	Role / Scope: Enabler; direct user breadth not adjudicated. Evidence Status: Partial	Direct AI deployment less well established; infrastructure significance high	Partial: direct AI breadth not adjudicated	Ch. 8; Table 8.3. Electricity and hardware concentration are consequential conditions, but the chapter identifies less direct AI evidence in critical infrastructure.
ANP-TW-DI-01	Defense & Intelligence	Role / Scope: User/enabler (public evidence partial). Evidence Status: Partial	Not adjudicated beyond limited public evidence	Partial: breadth not adjudicated	Ch. 8; Table 8.3. Public evidence is insufficient for a comparable defense/intelligence breadth judgment.

C.2.7 France

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-FR-CI-01	Commercial & Industrial	Role / Scope: Producer + user; enabling infrastructure. Observed Breadth: Substantial	Selected operational conversion; broader expansion underway	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Operational industrial use is established in selected sectors without implying uniform economy-wide maturity.
ANP-FR-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational shared national architecture	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Government AI use and shared administrative infrastructure are established; historical state claims are time-anchored separately.
ANP-FR-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Broad	Operational and very strong	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Public compute and research conversion provide broad science/education evidence.

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-FR-CR-01	Critical Infrastructure	Role / Scope: User (multiple selected subsectors). Observed Breadth: Substantial	Operational in transport and telecom; nuclear applications bounded	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Multiple critical sectors provide operational evidence, with nuclear AI treated conservatively.
ANP-FR-DI-01	Defense & Intelligence	Role / Scope: Producer + user/enabler (bounded public evidence). Observed Breadth: Substantial†	Selected operational support; force-wide integration not established	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Institutional/compute depth and support functions are visible; comprehensive defense/intelligence integration is not inferred.

C.2.8 Germany

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-DE-CI-01	Commercial & Industrial	Role / Scope: Producer + user. Observed Breadth: Broad	Very strong operational industrial conversion	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Industrial AI and manufacturing integration are broadly evidenced.
ANP-DE-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational; more integrated administrative architecture emerging	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Government use is established, while architecture continues to consolidate.
ANP-DE-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Broad	Operational and exceptional in scientific/industrial research	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Scientific compute and industrial research provide broad evidence.
ANP-DE-CR-01	Critical Infrastructure	Role / Scope: User (telecom established; other subsectors mixed). Observed Breadth: Selective	Operational telecom AI; other critical-infrastructure evidence remains mixed	Sufficient for selective breadth judgment	Ch. 9 §9.6; Table 9.3. Live telecom-network AI is established; the public record does not support a broader multi-subsector critical-infrastructure breadth claim.
ANP-DE-DI-01	Defense & Intelligence	Role / Scope: Producer + user/enabler (public evidence partial). Evidence Status: Partial	Development/testing/procurement strong; broad deployment not established	Partial: breadth not adjudicated	Ch. 9; Table 9.3. Defense technology activity is visible, but operational breadth remains insufficiently observable.

C.2.9 United Kingdom

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-UK-CI-01	Commercial & Industrial	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Operational ecosystem; less manufacturing-centered	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Commercial capability is strong through research, services, and firms rather than manufacturing depth alone.
ANP-UK-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational institutional adoption	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Public-sector AI use is strongly evidenced across institutions.
ANP-UK-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Broad	Operational and exceptional in research/public compute	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Research and public AI compute provide broad science/education capacity.
ANP-UK-CR-01	Critical Infrastructure	Role / Scope: User (selected functions). Observed Breadth: Selective	Telecom operational; electricity at live-trial/expanding stage	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Infrastructure maturity is selective and should not be generalized across all subsectors.
ANP-UK-DI-01	Defense & Intelligence	Role / Scope: Producer + user/enabler (bounded public evidence). Observed Breadth: Substantial†	Selected deployments/trials; broad integration not established	Sufficient for stated breadth judgment	Ch. 9; Table 9.3. Institutional mobilization is strong, but public evidence supports only selected defense/intelligence functions.

C.2.10 Russia

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-RU-CI-01	Commercial & Industrial	Role / Scope: Producer + user. Observed Breadth: Substantial	Operational in selected sectors; economy-wide breadth uneven	Sufficient for stated breadth judgment	Ch. 10 §10.11; Table 10.3. Major platforms and strategic firms sustain significant conversion under compute constraint.
ANP-RU-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Broad	Operational/high; maturity uneven across agencies and regions	Sufficient for stated breadth judgment	Ch. 10 §10.11; Table 10.3. Public-sector/platform integration is strong, but implementation is not uniform.
ANP-RU-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Operational/moderate under compute scarcity	Sufficient for stated breadth judgment	Ch. 10 §10.11; Table 10.3. Research capability persists despite high-end compute constraints.
ANP-RU-CR-01	Critical Infrastructure	Role / Scope: User + enabler. Observed Breadth: Substantial	Rail and telecom operational; other areas mixed	Sufficient for stated breadth judgment	Ch. 10 §10.11; Table 10.3. Incumbent infrastructure supports meaningful conversion, with maturity and automation authority uneven.
ANP-RU-DI-01	Defense & Intelligence	Role / Scope: Producer + user/enabler (public evidence partial). Evidence Status: Partial	Selected functions operational; breadth opaque	Partial: breadth not adjudicated	Ch. 10 §10.11; Table 10.3. Security-sector functions are observable, but classification prevents a comparable system-wide breadth judgment.

C.2.11 United Arab Emirates

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-AE-CI-01	Commercial & Industrial	Role / Scope: Enabler + user; producer role selective. Observed Breadth: Broad	Operationally diversified	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. Technology/cloud and major energy-sector applications establish broad commercial/industrial use.
ANP-AE-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Broad	Large-scale operational deployment	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. Government deployment and implementation architecture establish broad public-administration use.
ANP-AE-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Operational research ecosystem	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. TII, MBZUAI, and Falcon establish substantial research/model depth.
ANP-AE-CR-01	Critical Infrastructure	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational in selected strategic systems	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. Energy and sovereign digital infrastructure provide recurring operational evidence.

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-AE-DI-01	Defense & Intelligence	Role / Scope: Producer + user (defense-industrial evidence only). Evidence Status: Partial†	Defense-industrial AI operational; force-wide and intelligence-system integration not adjudicated	Partial: breadth not adjudicated	Ch. 11 §11.8; Table 11.3. EDGE evidence establishes defense-industrial AI, not armed-forces-wide or intelligence-system breadth.

C.2.12 Saudi Arabia

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-SA-CI-01	Commercial & Industrial	Role / Scope: Enabler + user. Observed Breadth: Broad	Operational strategic-industrial conversion	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. Aramco and other strategic-industrial applications support broad commercial/industrial evidence.
ANP-SA-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational institutional conversion	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. State data/AI institutions and HUMAIN-linked programs provide substantial public-sector conversion evidence.
ANP-SA-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Substantial and expanding; maturity mixed	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. University, training, model, and SDAIA-linked activity is substantial but still deepening.
ANP-SA-CR-01	Critical Infrastructure	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational in strategic industrial infrastructure; broader sectoral evidence uneven	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. Major strategic systems are well evidenced, but cross-sector breadth remains heterogeneous.
ANP-SA-DI-01	Defense & Intelligence	Role / Scope: Producer + user (defense-industrial evidence only). Evidence Status: Partial†	Defense-industrial AI established; force-wide maturity not established	Partial: breadth not adjudicated	Ch. 11; Table 11.3. Public evidence supports industrial capability but not a comparable breadth judgment for full defense/intelligence integration.

C.2.13 Qatar

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-QA-CI-01	Commercial & Industrial	Role / Scope: Enabler + user. Observed Breadth: Selective	Selective but operational	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. Sovereign cloud and LNG applications establish bounded commercial/industrial capacity.
ANP-QA-PA-01	Public Administration	Role / Scope: User + enabler. Observed Breadth: Selective	Operational in selected institutions; broader maturity uneven	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. Selected government functions are operational, but breadth across administration remains limited.
ANP-QA-SE-01	Science & Education	Role / Scope: Producer + enabler + user. Observed Breadth: Substantial	Strong research specialization; application maturity bounded	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. HBKU, QCRI, and Fanar provide strong relative research specialization.
ANP-QA-CR-01	Critical Infrastructure	Role / Scope: User + enabler. Observed Breadth: Substantial	Operational LNG, sovereign cloud, utility, and smart-meter functions	Sufficient for stated breadth judgment	Ch. 11; Table 11.3. Critical-infrastructure use is established across selected systems.

Claim ID	Function	Role / scope / breadth	Deployment maturity	Evidence sufficiency	Primary manuscript basis and residual limit
ANP-QA-DI-01	Defense & Intelligence	Role / Scope: Role not adjudicated from public evidence. Evidence Status: Insufficient	Not sufficiently established from public evidence	Insufficient for breadth judgment	Ch. 11; Table 11.3. The record is inadequate for a valid breadth judgment in defense and intelligence.

Functional-role qualifier. Country–function claims should also be read as producer, enabler, user, or mixed-role observations where that distinction matters. Semiconductor or infrastructure production is not treated as equivalent to downstream AI use; allocation, hosting, procurement, or research-compute provision can be an enabling contribution without establishing broad end-use deployment; and direct operational use is attributed only to the function in which it is observed. The five functional domains remain unchanged, but role differences constrain what breadth and maturity can mean within each cell.

C.3 Interpretation Rule

Read the register horizontally within a country–function claim and vertically only across commensurable analytical objects. Narrow but mature use, broad but heterogeneous use, and an unadjudicated classified-domain breadth are analytically distinct combinations, not points on a hidden rank. Supplement S4 should be consulted before treating any claim as source-complete: it identifies the principal evidentiary anchors and records where company self-report, dynamic documentation, secondary reporting, or public-evidence ceilings constrain inference.

Supplement S4 — Claim-to-Source Precision Audit

This appendix provides the source-level audit layer for the 65 Tier A ANP claim IDs in Supplement S3. The source anchors are deliberately selective rather than exhaustive: they identify the principal observations that make each claim auditable. Source role, timing, and evidentiary ceiling are recorded separately so that an official statistic, company deployment report, dynamic service page, secondary news account, or absence of public evidence is not silently treated as equivalent proof. Where a claim is coded Partial or Insufficient, the absence of a source capable of establishing breadth is itself part of the adjudication record. Company-reported performance establishes reported outcomes, not independently validated national effects; dynamic pages are used for cutoff-specific claims only where the relevant historical timing is stated or separately anchored.

D.1 Reading Rule

A source anchor in this appendix does not expand the claim beyond the wording in Supplement S3. The audit should be read from left to right: Claim ID → principal source anchor(s) → source role/timing → evidentiary ceiling. If the source role changes, or if a later source documents a different operational state, the time-indexed claim should be re-adjudicated rather than silently backfilled.

D.2 National Claim-to-Source Records

D.2.1 United States

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-US-CI-01	Grundy et al. (2026); Stanford HAI (2026)	Official U.S. Census adoption evidence + analytical ecosystem benchmark; dated 2026.	Establishes adoption/resource depth and broad commercial exposure, not uniform firm-level maturity or performance.
ANP-US-PA-01	GAO (2025, 2026); Rivera (2026)	Federal audit/inventory evidence + dated VA operational deployment.	GAO inventories mix maturity states; a large VA deployment does not establish uniform federal maturity.
ANP-US-SE-01	NSF (2026); DOE (2026b)	Official research-access and national-laboratory infrastructure records.	Establishes usable research access and institutional conversion, not research productivity or frontier superiority by itself.
ANP-US-CR-01	Berger (2026); T-Mobile US, Inc. (2026); Rivera (2026)	Operator/agency deployment records across grid, telecom, and federal health functions.	Selected-subsector evidence; operator performance claims are not national critical-infrastructure averages.
ANP-US-DI-01	U.S. Department of Defense (2024); U.S. Department of War (2026a, 2026b)	Official institutional and deployment records, including classified-network access architecture.	Public record establishes specified high-authority environments, not force/intelligence-wide breadth or mission effectiveness.

D.2.2 China

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-CN-CI-01	MIIT (2026a, 2026c, 2026d)	Official industrial-policy and manufacturing-adoption evidence.	Supports large-scale industrial conversion; ministry statistics and selected programs do not imply uniform firm-level capability.
ANP-CN-PA-01	CAC & NDRC (2025); Jilin Provincial Government Services and Digitalization Administration (2026); Jinan Municipal Big Data Bureau (2026)	National guidance + dated subnational operational cases.	Local operational cases establish real conversion but not uniform national administrative maturity.
ANP-CN-SE-01	NSFC (2026); Beijing Municipal Education, Science and Technology, and Talent Work Leading Group (2026)	Official science/AI-for-Science program evidence.	Establishes institutionalized research pathways, not uniform scientific output or frontier performance.
ANP-CN-CR-01	State Grid Corporation of China (2025); NEA (2026); China Telecom Corporation Limited (2026); MIIT (2026b)	Operator/regulator evidence across electricity, telecom, compute/network, and industrial infrastructure.	Operational support is established in several systems; autonomous authority and performance remain function-specific.
ANP-CN-DI-01	Zhao & Xiao (2026)	PLA Daily / Ministry of National Defense public operational-support case.	One public support-function case cannot adjudicate classified integration, combat availability, or intelligence-system breadth.

D.2.3 India

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-IN-CI-01	Ishwarbharath B & Suresh (2026); Monnappa (2026); MeitY (2026c)	Reuters reporting on GCC scale/capability + official human/technology program evidence.	GCC revenue and organizational scale are not direct measures of AI share or nationally controlled output.
ANP-IN-PA-01	MeitY (2026g, 2026d, 2026i, 2026h)	Official IndiaAI, BHASHINI, and public digital/compute program records.	Establishes institutional access and public-service conversion; program scale does not by itself measure service quality or benefit.
ANP-IN-SE-01	MeitY (2026h, 2026c); NITI Aayog (2026)	Official compute/talent/design and policy evidence.	Supports expanding scientific/educational capacity; does not establish frontier-scale compute parity.
ANP-IN-CR-01	PIB (2026)	Dated Government of India rail deployment record distinguishing operational systems from pilots.	Strong for the specified railway functions; does not establish broad critical-infrastructure maturity across all sectors.
ANP-IN-DI-01	DRDO (n.d.-a, n.d.-b); Ministry of Defence (2024, 2025)	Official technology-task pages + defense AI governance/platform records.	Establishes institutional/technical base; dynamic task pages and public program records do not reveal classified deployment scale or mission performance. Archive IDs: ARCH-IN-DRDO-AIML-20261005; ARCH-IN-DRDO-AUTO-20261005.

D.2.4 Japan

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-JP-CI-01	IPA (2026); Hitachi (2026)	Official adoption assessment + company operational factory deployment.	Hitachi performance gains are company-reported; economy-wide transformation remains uneven.
ANP-JP-PA-01	Digital Agency (2026c, 2026b, 2026a, 2026d)	Dated government rollout, model trial, and disaster-response deployment records.	Establishes substantial operational administrative use; access/rollout is not a national productivity estimate.
ANP-JP-SE-01	AIST (2025); METI (2026)	Official public-compute and foundation-model support records.	Establishes usable research compute and organized model development, not frontier-model breadth equal to the largest ecosystems.
ANP-JP-CR-01	SoftBank Corp. (2026)	Company record of AI verification in base-station integration operations.	Supports selected operational/verification evidence only; not nationwide autonomous network operation.
ANP-JP-DI-01	No adequate public source identified in the reviewed corpus	Public-evidence ceiling rather than a positive breadth record.	Partial status must not be converted into a claim of low latent capability or broad operational integration.

D.2.5 South Korea

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-KR-CI-01	MOTIR (2026); Samsung Electronics (2026)	Official manufacturing-program evidence + company HBM4 production record.	Selected-site productivity results are reported outcomes, not a national productivity estimate.
ANP-KR-PA-01	MOIS (2026)	Official launch of a pan-government AI legal assistant for central/local officials.	Service availability establishes operational conversion; breadth of sustained use and outcome quality remain separate.
ANP-KR-SE-01	MSIT (2025, 2026); Kakao Corp. (2025)	Official model/compute program evidence + company infrastructure completion record.	International recognition and infrastructure access do not establish global frontier parity.
ANP-KR-CR-01	KEPCO (2025); SK Telecom (2026)	Operator evidence from electricity substations and live telecom operations.	Strong operational evidence, but deployment/performance metrics remain operator-reported.

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-KR-DI-01	No adequate public source identified in the reviewed corpus	Public-evidence ceiling.	No breadth inference is made for classified defense/intelligence integration.

D.2.6 Taiwan

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-TW-CI-01	MOEA (2026a, 2026b); TSMC (2026a)	Official manufacturing-production and manufacturing-adoption evidence + company semiconductor-production evidence.	Hardware depth and manufacturing diffusion are strong; survey definitions are not directly comparable across countries.
ANP-TW-PA-01	MODA (2025, 2026)	Official government AI platform and agency-use records.	Platform access and selected workflows establish operational conversion, not government-wide productivity.
ANP-TW-SE-01	NSTC (2026); TAIDE (2026)	Official public-compute and localized-model evidence.	Establishes research/HPC access and local adaptation; external base-model dependence remains material.
ANP-TW-CR-01	Energy Administration, Ministry of Economic Affairs (2026); NSTC (2026)	Official electricity-system planning + public-compute context.	These sources establish infrastructure conditions and concentration, not enough direct AI deployment evidence to adjudicate breadth; status therefore remains Partial.
ANP-TW-DI-01	No adequate public source identified in the reviewed corpus	Public-evidence ceiling.	No breadth inference is made for defense/intelligence integration.

D.2.7 France

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-FR-CI-01	Airbus (2026); GENCI (2026c)	Company implementation record + public-private training-to-production case (RATP).	Some announced Airbus use cases remain prospective; production evidence is selected rather than economy-wide.
ANP-FR-PA-01	DINUM (2026); DINUM (n.d.-a, n.d.-b)	Dated September usage statistics + dynamic service/architecture documentation.	Cutoff classification uses the dated September record; dynamic cumulative figures are current context only. Archive ID: ARCH-FR-DINUM-20261005.
ANP-FR-SE-01	GENCI (2026a, 2026b, 2026c); CEA (2026)	Official/public research-compute and interoperability records.	Establishes strong scientific infrastructure and access, not uniform downstream outcomes.
ANP-FR-CR-01	EDF (2026); GENCI (2026c)	Critical-sector program + RATP production application.	EDF explicitly excludes plant-control autonomy; selected transport/nuclear-support cases do not imply uniform critical-infrastructure maturity.
ANP-FR-DI-01	Ministere des Armees (2025); Senat (2026)	Official classified-compute record + parliamentary review of defense-AI integration.	Supports selected defense functions and institutional depth; Senate evidence also limits claims of force-wide maturity.

D.2.8 Germany

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-DE-CI-01	Bosch (n.d.-a, n.d.-b); Federal Ministry for Economic Affairs and Energy (2026)	Company operational manufacturing documentation + dated federal industrial-AI initiative.	Bosch outcomes are company-reported; dynamic pages are used where they explicitly date deployments, not as undated cutoff snapshots. Archive IDs: ARCH-DE-BOSCH-SHOPFLOOR-20261005; ARCH-DE-BOSCH-DRESDEN-20261005.
ANP-DE-PA-01	SAP (2026); Federal Government of Germany (2026)	Dated sensitive-cloud authorization + federal-government public-service AI account.	Establishes selected operational public-sector functions; a unified government-wide AI service was still emerging.

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-DE-SE-01	Forschungszentrum Julich (2026); JUPITER access documentation cited in Ch. 9	Research-infrastructure and early-access evidence with dated operational anchors.	Early access does not equal fully routine production for the entire physical system.
ANP-DE-CR-01	Deutsche Telekom (2026)	Dated live-network RAN Guardian deployment.	Strong telecom evidence supports a Selective breadth judgment; the chapter does not establish comparable breadth across multiple critical-infrastructure subsectors.
ANP-DE-DI-01	Siebold (2026); Reuters (2026a)	Contemporary reporting on army AI priorities and future drone procurement.	Development/testing/procurement architecture is visible; broad operational deployment was not established by the cutoff.

D.2.9 United Kingdom

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-UK-CI-01	Wayve (2026); Sovereign AI (2026)	Company/government partnership + public compute allocation to UK AI companies.	Trials/partnerships and compute access do not establish nationwide embodied-AI deployment or uniform commercial maturity.
ANP-UK-PA-01	UK Government (2025b, 2025a)	Official operational trials and government-developed tool records.	Trial time savings are not a national productivity estimate.
ANP-UK-SE-01	Sovereign AI (2026); University of Bristol (2026a, 2026b); UK Government (2026)	Official/public research-compute and university infrastructure evidence.	Establishes exceptional research access/depth, not ownership of every frontier layer.
ANP-UK-CR-01	BT Group (2026); National Grid (2026)	Telecom operational integration + live electricity-demand trial.	Electricity evidence is a live trial, not grid-wide deployment; telecom uses an externally supplied frontier model.
ANP-UK-DI-01	UK Ministry of Defence (2026)	Official institutional mobilization and selected deployment/trial evidence.	Does not establish comprehensive force/intelligence integration or mission effectiveness.

D.2.10 Russia

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-RU-CI-01	Sber (2026); Yandex (2025, 2026); Rosneft (2026)	Domestic platform + strategic-industry operational records.	Evidence is concentrated in major incumbents; economy-wide breadth remains uneven.
ANP-RU-PA-01	Government of the Russian Federation (2025, 2026a, 2026c)	Official national/regional coordination and implementation records.	Government-reported regional counts require caution; maturity varies across agencies/regions.
ANP-RU-SE-01	HSE University (2026)	Institutional supercomputing/research record.	Establishes continuing research capability while simultaneously documenting compute scarcity.
ANP-RU-CR-01	Russian Railways (2025b, 2025a); T2 (2026); Rosneft (2026)	Operator/company evidence across rail, telecom, and energy/industrial systems.	Selected incumbent evidence supports substantial conversion, not uniform automation authority across all critical infrastructure.
ANP-RU-DI-01	Rostec (2026); Reuters (2026g)	Defense-industrial/public reporting on specified AI-enabled functions.	Classified breadth and system-wide operational integration remain opaque.

D.2.11 United Arab Emirates

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-AE-CI-01	ADNOC (2026); Core42 (2026)	Company operational energy deployment + sovereign compute availability.	ADNOC performance gains are company-reported; external upstream dependence remains material.
ANP-AE-PA-01	Abu Dhabi Media Office (2026)	Official rollout across 27 government entities.	Rollout/licence coverage establishes broad authorized deployment, not 35,000 active users or productivity gains.
ANP-AE-SE-01	TII (2026)	Institutional model/research record; supplemented by the chapter's MBZUAI/TII ecosystem evidence.	Strong dedicated research depth is established; cross-institutional output comparability remains limited.

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-AE-CR-01	Core42 (2026); ADNOC (2026); Reuters (2026l)	Operational sovereign compute/energy use + dated post-shock infrastructure adaptation.	Adaptation after a real shock is not proof of continuity under a future comparable disruption.
ANP-AE-DI-01	EDGE Group (2026)	Defense-industry manufacturing transformation record.	Establishes defense-industrial AI only; force-wide and intelligence-system breadth are not adjudicated, so status is Partial.

D.2.12 Saudi Arabia

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-SA-CI-01	Aramco (2025a, 2025b); AMD (2026)	Permanent industrial AI adoption + live domestic accelerator infrastructure.	Aramco benefits are company-reported; future MW/GW expansions are not counted as current capacity.
ANP-SA-PA-01	Saudi Press Agency (2026a); SDAIA public-service architecture described in Ch. 11	Dated Hajj Mobile Operations Centers with AI/real-time analytics supporting multiple agencies.	Establishes bounded cross-agency operational use, not uniform government-wide maturity.
ANP-SA-SE-01	Saudi Press Agency (2026b); SDAIA institutional/training architecture described in Ch. 11	Dated national capability-building scale plus institutional programs.	Training reach establishes human-capability mobilization, not independent research concentration or frontier-model performance.
ANP-SA-CR-01	Aramco (2025a, 2025b); AMD (2026)	Operational strategic-industrial systems + production compute.	Strong in strategic infrastructure; evidence remains uneven outside major state-linked industrial systems.
ANP-SA-DI-01	SAMI Advanced Electronics (2026)	Domestic defense-industry AI surveillance platform record.	Establishes defense-industrial capability, not force-wide or intelligence-system operational breadth.

D.2.13 Qatar

Claim ID	Principal source anchor(s)	Source role / timing	Evidentiary ceiling
ANP-QA-CI-01	QatarEnergy LNG (2025); Ooredoo Group (2026a, 2026b)	Industrial ML + operational sovereign cloud/data-center evidence.	Selective operational evidence; infrastructure scale is not equivalent to AI workload utilization.
ANP-QA-PA-01	National Planning Council (2026)	Official institutional Copilot adoption.	Supports selected institutional conversion only, not government-wide breadth.
ANP-QA-SE-01	HBKU (2026)	Institutional Arabic-model/research evidence.	Supports strong relative specialization; does not establish general frontier-model parity.
ANP-QA-CR-01	QatarEnergy LNG (2025); KAHRAMAA (2025, n.d.); Oracle (2026)	Direct industrial ML + dated smart-meter operations + sovereign cloud infrastructure.	Current dynamic meter counts are not backfilled into the cutoff; digital infrastructure is not automatically AI-specific. Archive ID: ARCH-QA-KAHRAMAA-SMARTMETERS-20261005.
ANP-QA-DI-01	No adequate public source identified in the reviewed corpus	Public-evidence ceiling.	Insufficient status records non-adjudication of breadth rather than zero capability.

D.3 Audit Interpretation

The audit reveals three recurrent evidence ceilings. First, company and operator records are often the best evidence of whether a system was actually deployed, but their reported performance effects are not treated as independently validated national outcomes. Second, public evidence in defense and intelligence is frequently sufficient to establish institutions, procurement, or selected support functions but insufficient to adjudicate breadth; Partial and Insufficient therefore remain legitimate outcomes. Third, infrastructure significance is not automatically AI deployment. This distinction leads, in the current version, to a Partial critical-infrastructure status for Taiwan, a Selective rather than Substantial critical-infrastructure breadth judgment for Germany, and a Partial defense/intelligence status for the UAE. These changes reflect source-role discipline rather than a change in the underlying ontology.

Supplement S5 — Author-Conducted Adversarial Adjudication Stress Test

This appendix reports an author-conducted adversarial stress test of the highest-risk classifications in the manuscript. It is not an independent red-team validation and does not add a new score or evidentiary layer. For each claim, the audit states the strongest plausible challenge, identifies the evidence that still carries the judgment after that challenge is accepted as a live rival interpretation, and specifies the observation that would require downgrade or revision. A claim is retained only at the narrowest level that survives that challenge.

E.1 Reading Rule

The stress test is deliberately asymmetric toward disconfirmation. "Keep" does not mean proven beyond dispute; it means that the current classification remains the most defensible one after the strongest identified challenge is applied. "Keep with qualification" means the claim survives only in the bounded form stated in the manuscript. The failure condition is prospective and observable: if it is met by better evidence, the time-indexed classification should change without altering the underlying construct definition.

E.2 High-Risk Claim Stress-Test Register

Claim / Object	Current Judgment	Strongest Challenge	Evidence That Survives the Challenge	Failure / Revision Condition	Disposition
DOMAIN-US-01	SUPPORTED non-exclusive U.S.-Centered AI Domain	Separate foreign embeddings may not constitute repeated same-function cross-layer organization. Technological origin, operating authority, and NATO accreditation must not be collapsed into U.S. control.	The Saudi production infrastructure contains a documented TS-C/TS-N bridge within one AI-infrastructure function. Other cases support bounded external functional recurrence at baseline, but do not supply additional strict-test passes. See DOMAIN-US-01 in Supplement S2.	Downgrade the baseline if qualifying external recurrence reduces to announcements or unrelated isolated cases. Upgrade the strict test only when independently documented, repeated same-function bridges satisfy the stated layer and attribution rules.	KEEP BASELINE WITH SCOPE LIMIT. STRICT TEST PARTIALLY SUPPORTED. Retain the non-exclusive domain judgment without claiming repeated strict-test success.
DOMAIN-CN-01	PARTIALLY SUPPORTED China-Centered External AI Configuration	Very large downloads and API-marketplace activity could be read as evidence that a China-centered external domain already exists.	The evidence establishes reach, but foreign use is often self-hosted, multi-homed, or mixed with U.S./European services; common cross-layer cloud–model–infrastructure–institutional organization remains incomplete.	Upgrade only if longitudinal evidence shows independently governed foreign actors recurrently organizing multiple AI functions around a common China-centered external architecture.	KEEP PARTIAL.
DOMAIN-EU-01	SUPPORTED EU AI-Relevant Regulatory Domain	The result may simply redescribe EU jurisdiction rather than identify a higher-order domain.	Repeated implemented supervision, procurement, market-access, data-use, and digital-service conditions recur across member states, regulators, firms, and services; the organizing logic is rule-based rather than technological.	Downgrade if the record reduces to formal legal scope without recurrent implementation across a meaningful actor set, or if the claimed differentiation is not operationally consequential.	KEEP WITH SCOPE LIMIT.
SI-MOSIP-01	Not Adjudicated; adjacent DPI calibration case	Mature implementations and project-level adoption totals risk defining the population by successful outcomes.	Adopter-side PhilSys and Fayda evidence supports implemented national identity-system change, continuing use, and traceable architecture lineage in selected cases.	Assign Class 0–3 only after the cutoff-date engagement cohort is independently enumerated by stage and system-relative breadth is actually tested.	DOWNGRADE FROM CLASS 3 TO NOT ADJUDICATED; retain methodological calibration.
SI-ADVCOMP-01	Not Adjudicated; persistent	The original Class 2 inference	U.S. regulatory controls and	Upgrade to Class 2 only after	RECALIBRATE TO NOT

Claim / Object	Current Judgment	Strongest Challenge	Evidence That Survives the Challenge	Failure / Revision Condition	Disposition
	cross-interface changes retained	risks selecting its functional universe from already-observed control episodes, without a sufficiently independent comparison population or justified system-relative breadth.	implementation, Dutch license revocation, Cadence delivery effects and rescission, and product-specific NVIDIA transactions establish material and repeated boundary changes. They do not, by themselves, demonstrate representativeness within the wider population.	comparing affected, unaffected, permitted, restored, and alternative relations and documenting consequential system-relative breadth; Class 3 additionally requires aggregate attribution. An adequate boundedness test could instead support Class 1.	ADJUDICATED (v48); retain observed cases and separate domain judgments.
SI-DPC-META-01	Not Adjudicated for provider-class SI; bounded persistent supervisory change established.	The broader DPC supervisory record and other EU-side adaptations might be used to infer cross-provider diffusion without establishing that the specific Meta-related structural effect recurred within an assessable comparator population.	The Meta relationship supports implemented and persistent change, while reported engagement with other AI products is relevant context rather than evidence of the same causal outcome across independent providers.	A defensible Class 1 judgment requires adequate system-relative comparison showing bounded reach; broader class judgments require independent recurrence and appropriate causal support. If these become available, initiate a new adjudication version.	RECLASSIFY: NOT ADJUDICATED (system-level).
ABP-APPLE-CN-01	State 2 operational boundary control + observable State 3 target adaptation	No named denial or suspension order is public; Apple's delayed China launch could reflect product readiness rather than public control.	Apple explicitly linked unavailability in China to regulatory requirements; filing, local-partner restructuring, and China-specific product architecture provide target-specific implementation and response evidence.	Downgrade to State 1 if credible evidence shows the launch condition was not operationally tied to the public admission architecture, or that technical/product readiness independently explains the China-specific non-launch.	KEEP; controller attribution bounded to the public admission architecture, not a named denial.
ABP-ANTH-DOD-01	Active State 2 at Pentagon military-contract interface	Litigation involved multiple legal tracks, so a broad "Anthropic blacklisting" claim could overstate the operative boundary.	The 25 September D.C. Circuit ruling sustained the supply-chain-risk designation and associated military-contract exclusion at the specified interface; the manuscript separately limits broader federal claims.	Downgrade if the exclusion was stayed, vacated, or shown not to govern the specified military-contract interface at the cutoff.	KEEP INTERFACE-SPECIFIC STATE 2.
BA-RUS-COMPUTE-01	Adaptation observed; BA gain insufficiently established	Switching from Western suppliers to Chinese or other foreign suppliers can itself increase autonomy relative to the original controller; continued foreign dependence is not disqualifying.	The manuscript now accepts that principle. What remains unproven is operational equivalence, sustained workload continuity, acceptable switching cost, durability, and independence from the original controlled interface.	Upgrade when a substitute path is shown to carry the specified function operationally and durably outside the original boundary at acceptable cost/performance.	KEEP UNESTABLISHED; foreign sourcing is not the reason.
BA-IMOD-CLOUD-01	Alternative path identified; realized BA gain unestablished	Reported movement toward a competing cloud may already show meaningful substitution.	Planning, contracting, or migration start demonstrates target response but not completed migration, equivalent workload performance, or sustained	Upgrade when the alternative is in actual operation with demonstrated functional continuity and reduced reliance on the original control interface.	KEEP UNESTABLISHED.

Claim / Object	Current Judgment	Strongest Challenge	Evidence That Survives the Challenge	Failure / Revision Condition	Disposition
			operation of the specified function.		
ANP-UAE-RES-01	Stress-responsive adaptation established; severe-disruption resilience unestablished	Post-attack redesign toward dispersion and hardening might be treated as proof of resilience.	Observed redesign is evidence of adaptation under a real shock, but the record does not demonstrate continuity of equivalent AI functions through a later disruption of comparable or greater severity.	Upgrade to resilience only with observed continuity, recovery, or observed stress performance under the relevant disruption class.	KEEP ADAPTATION / NOT RESILIENCE.
M1-CAP-ABP-01	Capacity-enabled boundary control; enabling mechanism, not causal ANP→ABP law	The same export-control episode could be double-counted as proof of national capacity and proof that national capacity caused external control.	The revised mechanism requires separately established ANP-relevant capacities/conversion pathways plus a distinct control right and implementation chain; it no longer treats aggregate ANP as a sufficient cause of ABP.	Mechanism remains unestablished where only formal boundary authority is observed or where no independently established capacity/conversion pathway is connected to implementation.	KEEP AS ENABLING CONFIGURATION; DO NOT CLAIM IDENTIFIED CAUSAL EFFECT.

E.3 Adversarial Review Result

The adversarial audit does not require a substantive upgrade of any high-risk claim. It does require one evidentiary repair and one vocabulary correction. First, the U.S.-centered domain discussion now relies on concrete foreign embeddings rather than a hypothetical composite user. Second, domain outcomes are described as supported, partially supported, not established, or rejected; "validated domain" is avoided because the present analysis provides a bounded, revisable evidentiary judgment rather than independent validation. A supported domain remains subject to the stated scope, observation window, and evidentiary limitations. The remaining high-risk claims survive only in their bounded forms: MOSIP remains an adjacent digital-infrastructure calibration case whose Systemic Influence classification is Not Adjudicated pending an independently enumerated comparison population; Apple and Anthropic remain interface-specific State 2 relations; Russia and IMOD retain unestablished Boundary-Autonomy gains; and UAE evidence establishes adaptation rather than resilience. The specified downgrade and upgrade conditions should be used in later time-indexed revisions rather than silently backfilling stronger interpretations.

E.4 Executed Qualitative Sensitivity Checks

U.S.-centered domain. The executed strict re-read excludes global prominence, single-layer foreign use, and cases where a non-U.S. authority is silently counted as a U.S.-origin layer. For this sensitivity test only, TS-C denotes accelerator/processor compute and TS-N denotes networking/fabric; these technical-stack codes are not the ANP functional categories. Saudi Arabia supplies the clearest pass: live AMD Instinct/EPYC compute (TS-C) and Cisco Silicon One networking (TS-N) operate inside one HUMAIN production AI-infrastructure function (Advanced Micro Devices, Inc., 2026). NATO Maven remains relevant as platform-to-governed-network embedding, but NATO owns the security accreditation and the supported model set is multi-origin (NATO JWC, 2026); Qatar is single-layer and UAE same-workload integration remains unestablished. The baseline domain remains Supported, while the strict same-function/two-U.S.-origin-layer sensitivity is Partially Supported because only one direct operational bridge clearly passes the stricter coding rule.

China Boundary Autonomy. The executed strict re-read removes announced orders, shipment totals, prospective collaborations, and generic substitution capacity from the constitutive BA evidence. Retained observations are task-level technical records: DeepSeek-AI (2026; v1) reports V4 execution validation on both NVIDIA and Ascend; Li et al. (2026; v3) report full-parameter V4-family post-training on Ascend SuperPOD with maintained training stability; and Yu (2026; v1) reports two V4-family inference workloads served correctly on 16 Ascend 910 devices while documenting substantial patching, throughput, and reliability constraints. These observations retain a bounded finding of operational alternative formation and task-level substitutability. Because the frozen record lacks a same-function earlier comparison point, the stricter re-read does not separately claim a temporal Boundary-Autonomy gain.

MOSIP population and Class 3 calibration. The stricter review separates the comparison population from successful outcomes. Project-reported adoption totals are not used as the population definition, and continuing operation or traceable lineage is not an eligibility condition. Under the current corpus, mature Philippine and Ethiopian implementations establish persistence and architecture lineage in selected cases, but the full cutoff-date cohort of country-level MOSIP engagements has not yet been independently enumerated by implementation stage. The Class 3 claim is therefore not retained in this edition: the MOSIP Systemic Influence candidate is Not Adjudicated pending a population-level stage register. Under the general SI logic, structural change or persistence failure yields Class 0; persistence with insufficient breadth yields Class 1; persistence and sufficient breadth with unresolved attribution yields Class 2; all gates with credible attribution yield Class 3; and inability to conduct a valid assessment yields Not Adjudicated. A broader comparison universe or weaker lineage evidence does not automatically imply Class 2.

ANP breadth labels. For borderline country–function cells, a stricter breadth rule treats several visible exemplars as insufficient for Broad. Broad requires recurrence across multiple major components of the specified functional universe; Substantial requires a nationally consequential but incomplete portion; Selective remains appropriate for narrow or specialized operating functions. Reclassifications under this stricter reading are recorded in Supplement S3 rather than absorbed into an aggregate score.

Supplement S6 — ASEAN Eleven-Member Node and Overlap Register

This appendix records the evidence basis for the ASEAN regional module in Chapter 21. It does not promote ASEAN to a national ANP actor and does not add the member states to the Tier A comparative matrix in Chapter 12. Each record identifies a primary and secondary functional node role, the observed national conversion base, principal external embedding, critical interfaces, target-side alternatives, contribution to ASEAN-level coordination, and an explicit evidence ceiling. Roles are descriptive, non-exclusive, and time-indexed to the common cutoff of 30 September 2026. Timor-Leste is included because it became ASEAN's eleventh member in October 2025.

F.1 Node-Role Reading Rule

A node role is not a rank. "Gateway," "industrial," "market," "security," "resource," and "emerging institutional" labels identify the function through which a state is most consequential to the regional AI environment in the observed period. The same state may possess several roles. Four evidence states qualify these labels: Supported, where current operation or established institutional coordination supports the role; Bounded/Function-Specific, where the role is demonstrated only for a narrow function; Formation-Stage, where strategy, investment, or institutional architecture is established but the proposed regional function is not yet fully operational; and Evidence-Constrained, where the public record is too thin for a stronger classification. External embedding can enable current capacity while also creating exposure to provider, supply, jurisdictional, or continuity boundaries. Boundary Autonomy concerns demonstrated capacity to create, activate, expand, or improve functionally credible alternatives to a specified external interface; an observed gain requires longitudinal evidence of improvement. Strategic intent, additional partnerships, or supplier diversity alone are insufficient. ASEAN-level coordination may alter national option sets, but this appendix does not treat "collective Boundary Autonomy" as a regional stock. To reduce repetition in the country cards, the third column uses the following shared guardrail codes: F-G1 = role labels are non-ordinal and non-exclusive; F-G2 = plans are excluded from current ANP; F-G3 = external embedding is not ownership or control; F-G4 = identifying an interface does not establish ABP; F-G5 = Boundary Autonomy requires a function-specific operationally credible alternative; F-G6 = regional contribution does not imply an ASEAN-wide controller; F-G7 = cutoff-bounded non-observation is not empirical absence; F-G8 = trajectory is separated from current classification; F-G9 = source roles are claim-specific.

A companion policy brief published on 11 October 2026 synthesizes the eleven-member configurations and regional overlap mechanisms (Wu, 2026h). It is included as related research, not as retrospective evidence for empirical classifications fixed at 30 September 2026.

F.2 Singapore

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Regional Gateway and Coordination Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Security / Interoperability Interface Node	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	NAIS 2.0 update, March 2026 launch of the National AI Impact Programme, national AI-governance tooling,high-performance-compute expansion, dense cloud/data-centre and enterprise ecosystem; proposed Digital Infrastructure Bill treated as institutional development at the cutoff.	F-G2 — plans excluded from current ANP.
External Technological Embedding	Multi-provider global cloud/model/accelerator ecosystem; strong links to U.S. technology firms and extensive regional/international connectivity.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Cloud/data-centre security and resilience; AI governance and assurance; cross-border payments/data; trusted digital infrastructure.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	Provider diversity suggests current choices, but no interface-specific Boundary-Autonomy gain is established; dependence on imported frontier hardware and global clouds remains material.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Supplies regional governance templates, interoperability practice, capital/services, and a trusted gateway for external technology networks.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Supported role. High public evidence of domestic strategy, governance, current infrastructure practice and deployment; the proposed Digital Infrastructure Bill is not coded as an operational licensing regime at the cutoff, and the record does not establish ASEAN-wide control.	F-G7 — cutoff-bounded; non-observation ≠ absence.

Field	Adjudication	Evidence / Interpretation
2030 Trajectory Test	By 2030, significance depends on sustaining trusted gateway functions while managing energy/land constraints and retaining multi-provider interoperability.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Smart Nation Singapore (2026); IMDA (2026a, 2026b, 2026c); MDDI (2026a, 2026b); ASEAN Secretariat (2024, 2026a).	F-G9 — source roles are claim-specific.

F.3 Malaysia

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Industrial–Compute Conversion Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Regional Hosting and Supply-Chain Gateway	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	Rapid data-centre expansion, electronics/semiconductor base, AI Malaysia Berhad formally established on 28 July 2026 with an inaugural board announced on 11 September, and policy to move from hosting toward AI factories, GPU services and local value capture.	F-G2 — plans excluded from current ANP.
External Technological Embedding	Global hyperscalers and accelerator ecosystems; foreign AI compute demand including China-linked users; locally governed transit and trade interfaces.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	AI-chip transit and export permitting; data-centre energy/water approvals; hosting continuity; local supply-chain participation.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	Multiple providers and domestic jurisdiction offer potential options, not verified substitutes to a specified upstream gate; frontier compute remains externally sourced and infrastructure constraints can limit substitution.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Adds regional compute/hosting capacity and creates a jurisdictional bridge between global chip controls, foreign AI demand, and Southeast Asian infrastructure.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Supported role. Strong infrastructure, institutional, and selected boundary-control evidence; performance-weighted compute and national access to all hosted capacity cannot be inferred.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	Trajectory depends on converting data-centre scale into domestic AI services, skills, semiconductor spillovers and sustainable infrastructure rather than remaining a hosting enclave.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Ministry of Digital Malaysia (2026a, 2026b); AI Malaysia Berhad (2026); MIDA (2026); Ministry of Investment, Trade and Industry Malaysia (2025); Reuters (2026b, 2026f).	F-G9 — source roles are claim-specific. Archive ID: ARCH-MY-AIM-FAQ-20261005 for the dynamic FAQ; dated Ministry of Digital releases remain the cutoff-timing anchors.

F.4 Indonesia

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Scale–Market and Platform Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Sovereign-AI Formation Node	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	Large domestic digital market and platform base; government policy stresses compute clusters, local models, talent and sovereign-AI capability formation. The strongest evidence at the cutoff concerns scale, demand and policy direction rather than realized upstream control.	F-G2 — plans excluded from current ANP.
External Technological Embedding	Collaborative multi-source technology environment; government explicitly rejects a simple user-only role and describes a middle path in global technology competition.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Compute localization and cloud access; data and platform governance; local model/talent development; energy and infrastructure for compute.	F-G4 — interface identification ≠ ABP.

Field	Adjudication	Evidence / Interpretation
Boundary Autonomy Assessment	Market scale and a collaborative multi-source strategy may enlarge future options, but operational equivalence of domestic alternatives is not established across the upstream stack; no broad Boundary-Autonomy gain is coded.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Provides demand scale, data and application markets, and a major test of whether regional market power can be converted into domestic capability.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Formation-stage for sovereign-AI role. Market/platform scale and policy direction are clear; realized national high-end compute control and broad autonomous substitution remain unestablished.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	2030 trajectory hinges on whether planned compute clusters, talent and local models become routinely usable national capabilities rather than policy aspirations.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Ministry of Communication and Digital Affairs Indonesia (2026a, 2026b).	F-G9 — source roles are claim-specific.

F.5 Viet Nam

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Industrial–Compute Conversion Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Scale–Market and Data Node	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	AI strategy approved on 28 August 2026; FPT AI Factory's reported operating services and expansion plans; manufacturing, digital services and localized model initiatives.	F-G2 — plans excluded from current ANP.
External Technological Embedding	Deep reliance on NVIDIA-class accelerators and globally distributed semiconductor/cloud supply chains alongside domestic systems integration.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Accelerator access, cloud/AI-factory services, data localization and local-language model development, industrial digitalization.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	Domestic firms operate parts of the software and service stack, but no longitudinal comparison establishes a Boundary-Autonomy gain; frontier-hardware substitutes remain limited.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Adds regional AI-factory capacity, software services, manufacturing depth, localized data/model resources, and a bridge to Northeast Asian/global supply chains.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Supported within the stated industrial-compute scope. The evidence concerns identified firm-level deployment and service activity, including the FPT/NVIDIA-linked pathway; it does not establish unrestricted national mobilizability, resilience under a severe external-access shock, or indigenous frontier-hardware production. The national inference therefore remains narrower than the observed corporate operating contribution.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	Could deepen if external hardware is converted into sustained regional services and local model development.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Government of Viet Nam (2026); FPT AI Factory (2026).	F-G9 — source roles are claim-specific.

F.6 Philippines

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Scale–Market and Data Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Strategic-Security Interface Formation Node	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	National AI roadmap, research institutions and digital-services market; EO 119 sets phased government-data classification and residency requirements; the proposed Pax Silica hub is not	F-G2 — plans excluded from current ANP.

Field	Adjudication	Evidence / Interpretation
	operational.	
External Technological Embedding	Pax Silica participation and proposed U.S.-linked industrial cooperation; separate investment and energy partnerships with Gulf actors and other providers.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Government data residency; industrial-hub access; semiconductor/critical-mineral and advanced-manufacturing links; cloud and security partnerships.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	Data-governance authority and partnerships may preserve certain choices, but credible alternatives at specified AI interfaces remain unestablished; the proposed AI-native industrial hub remains prospective.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Provides a security-industrial bridge linking ASEAN markets to U.S.-led technology networks while retaining broader investment relationships.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	The scale-market role and issued data-governance rules are Supported within scope; EO 119 provides for phased implementation, so full agency compliance is unverified. The proposed AI-native hub remains Formation-Stage, not current operating capacity. “Mixed” combines role-specific statuses rather than constituting a separate category.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	Trajectory depends on whether Pax Silica and data-centre investments become operational domestic value chains rather than externally anchored projects.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Department of Trade and Industry Philippines (2024); Board of Investments Philippines (2026); Philippine Information Agency (2026).	F-G9 — source roles are claim-specific.

F.7 Thailand

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Cloud–Industrial Conversion Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Regional Hosting Gateway	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	Operational cloud regions, major data-centre investment, deep manufacturing/industrial base, public AI/data-centre policy and Thai-language/LLM initiatives.	F-G2 — plans excluded from current ANP.
External Technological Embedding	Multi-hyperscaler environment with major foreign investment; domestic policy oversight and evolving data-centre security, sustainability and sector-specific regulatory criteria.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Existing applicable approvals and proposed sector-specific data-centre criteria; energy and water use; cloud/data residency; technology transfer; local talent and industrial integration.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	Provider diversity may expand choices, but no observed Boundary-Autonomy gain is established; frontier models, accelerators and clouds remain externally supplied.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Adds large-scale hosting, manufacturing and enterprise demand and offers an alternative Southeast Asian infrastructure location alongside Singapore and Malaysia.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Supported for hosting/cloud conversion. Operational cloud and policy evidence are strong; current AI-factory depth and domestic ownership of frontier layers remain more limited.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	Could become a larger regional hosting and industrial-AI node if infrastructure growth produces technology transfer and domestic	F-G8 — trajectory separated from current state.

Field	Adjudication	Evidence / Interpretation
	ecosystem depth.	
Principal Source Anchors	Bajwa (2026); Prakoso (2025); Royal Thai Government (2026).	F-G9 — source roles are claim-specific.

F.8 Brunei Darussalam

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Small-State Governance Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Sovereign-Infrastructure Formation Node	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	Digital Brunei 2030 integrates digital government, digital society, business and a cross-cutting Data and AI Strategy; it also identifies an AI-capable data centre, sovereign cloud with AI compute, and a Brunei-context national LLM as implementation priorities. These remain formation evidence unless operational status is separately demonstrated.	F-G2 — plans excluded from current ANP.
External Technological Embedding	Externally supplied technology is important, but the public corpus reviewed here does not support precise attribution of a frontier compute stack or dominant provider relationship.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Public digital systems, data governance, AI policy, procurement, cloud/compute formation and external technology partnerships.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	The proposed sovereign cloud and national model may expand future options, but credible substitutes at major AI interfaces and observed autonomy gains remain unestablished.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Provides a small-state test of whether concentrated resources and governance coordination can create specialized regional functions without large market scale.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Formation-stage. Strategy and coordinated implementation architecture are clear; frontier-scale operational AI depth, capital-energy conversion, and major-interface alternatives remain insufficiently evidenced.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	2030 trajectory depends on converting planning and resource advantages into live public, industrial and compute functions with traceable external dependencies.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Ministry of Transport and Infocommunications Brunei Darussalam (2026).	F-G9 — source roles are claim-specific.

F.9 Cambodia

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Emerging Institutional and Application Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Multi-Partner Institutional Orientation	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	Draft National AI Strategy and ongoing AI-governance preparation; digital-government programs; policy focus on human capital, infrastructure/data, public-service AI and responsible governance.	F-G2 — plans excluded from current ANP.
External Technological Embedding	Government articulates inclusive neutrality and seeks cooperation, technology transfer and investment across external partners. This establishes a multi-partner orientation, not operationally demonstrated multi-provider substitution.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Cloud-first/public digital infrastructure, data governance, AI policy, skills, model/compute access and external development partnerships.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	Multi-partner orientation can create future optionality only if alternative suppliers and infrastructures become operationally credible; current evidence is stronger for policy posture than for	F-G5 — BA capacity and temporal gain need distinct evidence.

Field	Adjudication	Evidence / Interpretation
	realized substitution.	
ASEAN-System Contribution	Contributes a development-oriented governance and application perspective and tests whether ASEAN interoperability can accommodate lower infrastructure depth.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Formation-stage. Policy and institutional formation are visible; compute scale, operational supplier substitution, model depth and continuity evidence remain limited.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	Trajectory depends on moving from draft strategy and governance to operational infrastructure, public-service deployment and locally anchored capability.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Hun Manet (2026); Ministry of Post and Telecommunications Cambodia (2026).	F-G9 — source roles are claim-specific.

F.10 Lao PDR

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Emerging Public-Digital Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Connectivity-Dependent Institutional Node	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	National AI strategy work; reported April 2026 launch of G-Chat v2 with LaoAI; public-administration digitization and workforce initiatives. Uptake remains unverified.	F-G2 — plans excluded from current ANP.
External Technological Embedding	Strong need for international technology and investment partners; domestic platforms coexist with wider dependence on external infrastructure, equipment and expertise.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Government communications, national data/digital platforms, cloud/connectivity, investment administration, cybersecurity and local-language AI.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	The reported LaoAI/G-Chat v2 launch identifies a public-sector AI interface, not proven upstream substitution or an observed Boundary-Autonomy gain.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Adds localized public-AI experimentation and a Mekong connectivity case to the regional coordination environment.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Bounded role. Platform launch is reported; uptake, reliability, compute depth and disruption resilience remain unverified.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	2030 trajectory depends on whether current public-sector platforms seed wider domestic skills and services while improving infrastructure access.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Lao News Agency (2026a, 2026b); Ministry of Technology and Communications Lao PDR (2026).	F-G9 — source roles are claim-specific.

F.11 Myanmar

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Evidence-Constrained Emerging Application Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Institutional-Formation Node	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	Public records document National AI Strategy/Policy drafting and an internally tested pilot of a legal-AI assistant; wider public deployment and effects remain unverified.	F-G2 — plans excluded from current ANP.
External Technological Embedding	The available public corpus does not support a reliable map of current cloud, accelerator, model or external-provider dependence across the national system.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Administrative AI, strategy formation, skills, digital-government applications and future governance rules.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	Not adjudicated at major interfaces. Additional suppliers or	F-G5 — BA capacity and temporal gain need distinct evidence.

Field	Adjudication	Evidence / Interpretation
	policy statements cannot be treated as credible alternatives without operational and function-specific evidence.	
ASEAN-System Contribution	Contributes evidence of policy drafting and a selected internal AI pilot, but cannot be assigned a stronger regional node role from the available public record.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Evidence-constrained. Public-evidence ceiling is material; claims are intentionally limited to the reported internal pilot and policy formation.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	Future classification should change only when operational infrastructure, recurring use and external-interface conditions become publicly observable.	F-G8 — trajectory separated from current state.
Principal Source Anchors	Ministry of Information Myanmar (2025, 2026).	F-G9 — source roles are claim-specific.

F.12 Timor-Leste

Field	Adjudication	Evidence / Interpretation
Primary Node Function	Regional Integration and Institutional-Formation Node	F-G1 — role code; non-ordinal/non-exclusive.
Secondary Node Function	Connectivity Formation Node	F-G1 — secondary role; non-exclusive.
Observed ANP / Conversion Base	Digital Transformation Strategy 2026–2036 under development with pillars for infrastructure, interoperable public services, digital skills, cybersecurity and regulation of emerging technologies including AI. The record primarily establishes institutional and capacity formation, not mature AI deployment.	F-G2 — plans excluded from current ANP.
External Technological Embedding	ASEAN membership creates coordination opportunities, not verified technical access; infrastructure and national institutional capacity remain developing.	F-G3 — embedding ≠ ownership/control.
Critical Interfaces	Regional interoperability, digital infrastructure, government modernization, cybersecurity, platform governance and AI regulation.	F-G4 — interface identification ≠ ABP.
Boundary Autonomy Assessment	Current autonomy is not established at major AI interfaces; the immediate analytical issue is whether diversified regional access can be created during capacity formation.	F-G5 — BA capacity and temporal gain need distinct evidence.
ASEAN-System Contribution	Tests whether ASEAN cooperation can lower coordination costs; membership does not establish interoperability or usable AI capacity.	F-G6 — regional contribution ≠ ASEAN-wide control.
Evidence Ceiling	Formation-stage. Strategic direction and ASEAN integration are observable; mature nationally usable AI deployment and major-interface Boundary Autonomy remain unestablished.	F-G7 — cutoff-bounded; non-observation ≠ absence.
2030 Trajectory Test	2030 trajectory depends on infrastructure, skills and institutional implementation, and on whether ASEAN coordination lowers the cost of accessing multiple external systems.	F-G8 — trajectory separated from current state.
Principal Source Anchors	ASEAN Secretariat (2025c); Government of Timor-Leste (2026).	F-G9 — source roles are claim-specific.

F.13 Regional Adjudication

The eleven records support four conclusions. First, ASEAN heterogeneity is structured rather than residual: different members contribute gateway, industrial, market, security, public-digital, governance, and institutional-formation functions. Second, the maturity of those roles differs materially. Singapore, Malaysia, Viet Nam, and Thailand have the strongest current operational or institutional anchors for their principal roles; Lao PDR provides a bounded function-specific public-digital case; Indonesia, the Philippines, Brunei, Cambodia, and Timor-Leste include important formation-stage roles whose future significance must not be back-cast into current capacity; and Myanmar remains evidence-constrained. Third, regional coordination is strongest where interoperability and common reference frameworks can operate without centralized enforcement. Fourth, multi-source external embedding can expand national option sets but does not automatically create Boundary Autonomy, and no ASEAN-wide stock of collective Boundary Autonomy is coded. The formal regional judgment

therefore remains unchanged: ASEAN Regional Coordination Environment — Institutionally Consequential; ASEAN AI Power Domain — Not Established. Function-specific intersections are assessed separately where the overlap test can be completed; the proposed integrated Malaysia–Singapore advanced-compute operational chain remains Not Adjudicated.

F.14 ASEAN Node-Role Adversarial Stress Test

This stress test asks whether each member-state role would survive the strongest plausible overclaim challenge. It does not create an additional score. A role is retained only at the level supported by current operation, established institutional coordination, or a clearly bounded formation pathway; planned infrastructure, market size, strategic alignment, or supplier multiplicity are not promoted into realized capacity, Boundary Autonomy, or regional control.

Member State	Strongest Overclaim Risk	Final Disposition	Upgrade / Downgrade Condition
Singapore	Regional gateway could be mistaken for ASEAN-wide control.	Keep Supported gateway/coordination role; no ASEAN-wide controller claim.	Downgrade only if regional coordination and domestic conversion cease to be recurrent or operational.
Malaysia	Data-centre FDI could be mistaken for Malaysian ownership or usable access to all hosted compute.	Keep Supported industrial/hosting role; locally administered transit control remains interface-specific.	Downgrade if hosting growth lacks domestic functional connection or selected interface control cannot be sustained.
Indonesia	Market size and sovereign-AI rhetoric could be mistaken for realized upstream conversion.	Retain scale-market/platform role; sovereign-AI role remains Formation-Stage.	Upgrade only with operational compute/model pathways; downgrade if current conversion beyond demand scale remains unobservable.
Viet Nam	Private FPT capacity could be generalized into national mobilizability.	Keep industrial-compute role with explicit private-capacity and hardware-dependence limits.	Downgrade if AI-factory services are not routinely operational or regionally usable.
Philippines	Pax Silica and the AI-native hub could be treated as present capacity or current strategic control.	Primary role moved to scale-market/data; strategic-security interface retained only as Formation-Stage.	Upgrade only after operational industrial/AI infrastructure and demonstrated national functional connection.
Thailand	Foreign cloud regions could be equated with domestic control of the frontier stack.	Keep cloud-industrial conversion role; provider diversity is not Boundary Autonomy by itself.	Downgrade if cloud/data-centre presence lacks sustained domestic functional conversion.
Brunei Darussalam	Hydrocarbon/resource wealth could be presumed to produce capital-energy AI conversion.	Remove capital-energy conversion claim; retain small-state governance and sovereign-infrastructure Formation-Stage role.	Upgrade only with live AI compute/cloud/model functions and traceable national access.
Cambodia	Inclusive neutrality could be mistaken for operational multi-alignment or Boundary Autonomy.	Retain institutional/application formation and multi-partner orientation only.	Upgrade only when alternative providers/infrastructures preserve specified functions in practice.
Lao PDR	LaoAI could be generalized from one public function to the national stack.	Keep Bounded/Function-Specific public-digital role.	Upgrade only with recurring cross-sector deployment and stronger infrastructure evidence.
Myanmar	Sparse official evidence could be overgeneralized into a national AI configuration.	Keep Evidence-Constrained status and narrow application/policy claims.	Any stronger role requires independently observable infrastructure, deployment, and interface evidence.
Timor-Leste	ASEAN membership and strategy formation could be mistaken for current AI capacity.	Retain regional integration/institutional-formation role only.	Upgrade only when planned digital/AI capabilities become operational and nationally connected.

Source-preservation note. This supplement reproduces the complete original Appendix B–F tables, plus the detailed targeted-review and dynamic-retrieval tables extracted from Appendix A. It is not an independent re-adjudication; consult the edition-specific release ledger before using it as evidence of subsequent changes.